

Driving License Management

Prepared by

Ahmad Asfar - Hasan Homsy - Adnan Masalmeh

Supervised by

Dr. kadan aljoumaa

Eng. Mohamad Albalkhi

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ABSTRACT

The “Driving License Management” project is considered a practical application of software engineering concepts, as it embodies the Software Development Life Cycle (SDLC), starting from requirements gathering and analysis, through design, implementation, and testing. The system aims to automate processes related to managing trainees’ data, organizing schedules, recording test results, and issuing and renewing driving licenses.

In this project, a systematic and well-planned development methodology was adopted by documenting functional and non-functional requirements, designing UML diagrams such as Use Case, Class Diagram, and ERD, and implementing the system using appropriate technologies such as C#, SQL Server, and Entity Framework.

The project contributes to reinforcing software engineering concepts in terms of systematic analysis, system design, and ensuring software quality through comprehensive test cases, reflecting a real application of academic standards in building reliable and scalable systems.

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Chapter 1 Introduction

1. Introduction:

In this chapter, we introduce the "Driving License Management System", a web application developed to streamline and automate the administrative processes in driving license centers. The traditional methods of handling license applications, exam scheduling, trainee records, and license issuance are often manual, error-prone, and time-consuming. These inefficiencies can lead to delays, mismanagement, and lack of centralized control.

2. Problem Definition:

Driving license centers currently suffer from heavy reliance on manual processes to manage trainee information, organize training schedules, record exam results, and issue or renew licenses. These traditional methods often result in lost data, human errors, delays in processing applications, and difficulties in tracking each trainee's status.

The lack of a unified system to manage these operations leads to data redundancy, lack of proper monitoring, and increased administrative workload. This negatively affects service quality and efficiency. Additionally, the absence of clear user role permissions poses a risk to system security and control.

With the growing demand for digital transformation and improved operational efficiency, there is a clear need for a comprehensive digital system that automates all processes related to driving license management in a secure, organized, and efficient manner. The system should provide a user-friendly interface for both administrators and operators.

3. Project objectives:

The primary objective of this project is to develop a software system that supports driving license centers in managing trainee and license-related operations electronically and efficiently. This system aims to improve operational performance, reduce human errors, and ensure that services align with user needs for a more seamless and productive experience.

To achieve this, the system will provide key functionalities such as adding and managing trainee data, organizing training schedules, recording exam results, issuing and renewing licenses, and controlling access based on user roles (admin/officer/citizen). The system will also generate detailed reports to assist in tracking processes and enhancing administrative oversight.

Ultimately, these objectives aim to automate essential operations, streamline information access, and shorten processing time—resulting in improved user satisfaction and better service quality at driving license centers.

4. Report organization:

The report will be organized in the following chapters:

- Chapter 1: introduction.
- Chapter 2: Basic concepts and literature review.
- Chapter 3: project management.
- Chapter 4: System analysis.
- Chapter 5: design study for the system
- Chapter 6: practical implementation.
- Chapter 7: Conclusion.

Chapter2

Basic concepts and literature review.

1. Introduction:

In this chapter, an overview of the main concepts, components, and tools used in the proposed Driving License Department (DVLD) system is presented. This chapter also includes a reference study of similar systems that are currently in use for managing license-related applications and services.

The purpose of this chapter is to establish a clear understanding of the theoretical and practical foundations upon which the system is built. By reviewing existing systems and identifying their key features and limitations, this study helps justify the design decisions adopted in the proposed system and ensures alignment with real-world requirements and best practices in information systems development.

2. Basic concepts:

2.1 Software Maintenance

Software maintenance in the DVLD system is essential to ensure continuous operation and compliance with changing regulations and administrative procedures. Maintenance activities include updating application workflows, modifying license rules, correcting system errors, and improving performance and security. Continuous maintenance ensures the system remains reliable for both citizens and employees while supporting long-term operational needs.

2.2 Requirements Engineering

Requirements engineering is the process of analyzing, defining, and documenting the system requirements for the DVLD system. This includes identifying all functional requirements such as managing people records, employee accounts, applications, tests, and licenses, as well as non-functional requirements like security, performance, and usability. Proper requirements engineering ensures that the system meets organizational goals and provides accurate and efficient digital services.

2.3 User Roles and Access Control

The DVLD system supports multiple user roles, including citizens, employees, and administrators. Each role has specific permissions and responsibilities within the system. Access control mechanisms ensure that users can only perform actions related to their role, such as submitting applications, managing tests, or approving licenses. This role-based approach enhances system security and operational integrity.

2.4 Application Management Systems

Application management is a core concept in the DVLD system. It includes creating, modifying, tracking, and canceling applications related to driving licenses and related services. The system maintains application statuses, payment information, and processing history, allowing both citizens and employees to track application progress efficiently and transparently.

2.5 Data Integrity and Validation

Data integrity is critical in managing personal information and official records. The DVLD system enforces validation rules to ensure correctness and consistency of data such as national numbers, dates of birth, license classes, and application types. This reduces errors and ensures accurate decision-making during application processing.

2.6 Workflow Management

The DVLD system follows predefined workflows for processing applications, conducting tests, and issuing licenses. Each workflow defines clear steps and conditions, such as eligibility checks, test scheduling, result recording, and final approval. Workflow management ensures consistency, traceability, and compliance with regulations.

2.7 Database Management Systems

The DVLD system relies on a structured relational database to store and manage data related to people, users, applications, licenses, and tests. Proper database design ensures efficient data retrieval, relationship management, and scalability. Entity relationships and junction tables are used to represent complex associations between system components.

2.8 Data Reporting and Monitoring

The system provides reporting and monitoring features to support administrative decisions. These reports help track application statuses, test outcomes, license issuance, and system usage. Accurate reporting enhances transparency and improves operational planning.

2.9 System Security

Security is a fundamental aspect of the DVLD system. It includes user authentication, password management, role-based authorization, and protection of sensitive personal data. Security mechanisms ensure that system access is controlled and that data confidentiality and integrity are maintained.

3. Literature review:

3-1) RTA – Road and Transport Authority (نظام هيئة الطرق و المواصلات)

Where is it applied?

Dubai Emirate – United Arab Emirates, and it is considered one of the most advanced systems in the Middle East.

Main functions:

- Creating an account for the applicant
- Submitting a license application electronically
- Uploading documents
- Linking the training center with the eye test center
- Theory test – parking test – road test
- Postponing or canceling tests
- Electronic payment
- Tracking application status
- Restrictions on submitting applications for certain categories



3-2) DMV – Department of Motor Vehicles (نظام إدارة المركبات)

Where is it applied?

Each state in the United States has its own DMV system.

Main functions:

- An electronic account for the applicant
- Online license application submission
- Vision test
- Theory test
- Driving test
- Rescheduling appointments
- Cancellation
- Tracking application status
- Clear instructions after failing the test



3-3) DVLA – Driver Vehicle Licensing Agency (نظام هيئة ترخيص السائقين)

The official government system for issuing driving licenses in the United Kingdom.

Main functions:

- Gov.UK user account
- Online application submission
- Fee payment
- Booking the theory test
- Booking the driving test after passing
- Changing appointments
- Tracking the application
- Receiving the result and the license



Driver & Vehicle
Licensing
Agency

Based on the study of the previous systems, the system requirements for our project were derived and include several main areas:

People Management

- Add / Edit / Delete / View / Search for a person

Employee Management

- Create an employee account
- Edit / Delete / View employee data / Search for an employee
- Change password

Application Management

- View applications (track application status)
- Submit a local driving license application
- Cancel / Delete / Edit applications / Search for an application

Test Management (Vision – Theory – Practical)

- Schedule each test
- Ability to modify test appointments
- Record test results
- Ability to reschedule tests in case of failure

Local Driving License Management

- Renewal / Replacement for damaged license / Replacement for lost license

International Driving License Management

Violation Management

- License suspension / Unsuspension

Application Type Management

Feature / System	Our System	بريطانيا DVLA	امريكا DVM	دبي RTA
Online application for a driving license	☒	☒	☒	☒
Vision test	☒	☒	☒	☒
Theory test	☒	☒	☒	☒
Practical driving test	☒	☒	☒	☒
Ability to modify or reschedule appointments	☒	☒	☒	☒
Ability to cancel the application	☒	☒	☒	☒
Track application status	☒	☒	☒	☒
Electronic payment	☒	☒	☒	☒
Sending notifications to the user	☒	☒	☒	☒
User and role management	☒	☒	☒	☒
User profile	☒	☒	☒	☒
Single sign-on system	☒	☒	☒	☒
Driving license management	☒	☒	☒	☒
International license management	☒	☒	☒	☒
Violation management	☒	☒	☒	☒

Chapter3 Project Management

1. Introduction:

In this chapter, the project management aspects of the Driving License Department (DVLD) system are presented. This chapter focuses on the planning, organization, and control activities required to ensure the successful development and delivery of the system.

It covers key project management components including the project charter, project plan, Statement of Work (SOW), stakeholder identification and analysis, as well as risk management strategies. These elements provide a structured framework to guide the project through all phases of development, from initiation to implementation, while ensuring alignment with project objectives, timeline, and quality standards.

2. Project charter:

A project charter is a formal document that serves as an official authorization for the start of a project. It acts as a reference point throughout the project, providing a clear understanding of the project's purpose and establishing a foundation for decision-making and project governance.

Project title: Driving License Management System

Project start date: October 18 2025

Project finish date: December 31 2025

Project manager: [Raghad Al-Hossny]

Project objectives: Develop a desktop-based software system to support driving license centers in managing trainee records, scheduling lessons, handling exam results, and issuing driving licenses efficiently.

Approach

1. Define project scope and objectives
2. Analyze user requirements and system features
3. Design system architecture and database structure
4. Develop core modules: trainee registration, scheduling, exams, and license issuance
5. Test and validate the systems functionality
6. Add reporting and print/export features
7. Perform user testing and gather feedback
8. Document all phases and outputs of the project

Roles and responsibilities

Name	Role	Responsibility
Dr. Tharaa Aslan	Supervisor	Project oversight, review, and guidance
Eng. Raghad Al-Hossny	Supervisor	Technical direction and monitoring
[Ahmad-Hasan-Adnan]	Engineer	Backend development, UI design, documentation

3. The SOW document:

3.1 Overview

The Statement of Work (SOW) defines the scope, objectives, deliverables, timeline, and responsibilities associated with the development of the Driving License Department Management System (DVLD). This document serves as a formal agreement that clarifies what the project will deliver and how it will be executed to ensure successful completion within the defined timeframe.

3.2 Project Title

Driving vehicle License Department Management System (DVLD)

3.3 Project Description and Objectives

The DVLD system is a web-based software solution designed to automate and manage the core operations of a driving license department. The system enables individuals to submit license-related requests electronically, track their application status, and receive notifications, while allowing employees and administrators to manage requests, tests, licenses, and violations efficiently.

The main objectives of the project are:

- To digitize and automate driving license services.
- To reduce manual paperwork and administrative workload.
- To improve accuracy, transparency, and efficiency in processing requests.
- To provide a centralized system for managing people, employees, requests, tests, and licenses.

3.4 Project Scope

The scope of the project includes the design and implementation of a management system that covers the following functional areas:

- Person and user account management.
- Employee management and role-based access.
- Electronic submission and tracking of license-related requests.
- Management of driving tests (vision, theoretical, and practical).
- Management of local and international driving licenses.
- Handling violations, license blocking, and unblocking.

- Management of request types and test types.

3.5 Project Goals

- Develop a reliable and user-friendly system for driving license services.
- Allow applicants to submit and follow up on their requests electronically.
- Enable employees to manage operational tasks efficiently.
- Ensure data consistency, security, and traceability.
- Apply software engineering best practices using Agile Scrum methodology.

3.6 Project Deliverables

The project deliverables include:

- Project Plan.
- Software Requirements Specification (SRS) document.
- UML diagrams (Use Case, Class, Sequence, Activity).
- Database schema and ERD.
- Fully functional backend and frontend system.
- Final project report.
- Presentation and system demonstration.

3.7 Technology and Tools

- **Programming Languages:** C#, JavaScript, HTML, CSS
- **Frameworks:** .NET (Backend and Front), Web-based UI technologies
- **Database:** SQL Server / MySQL
- **Modeling Tools:** ERDPlus, Draw.io
- **Methodology:** Agile Scrum

3.8 Assumptions

- Availability of project requirements and domain knowledge.
- Continuous access to supervisors for feedback and guidance.
- Commitment of all team members throughout the project duration.
- The system will be developed for academic and training purposes.

3.9 Project Resources

Human Resources:

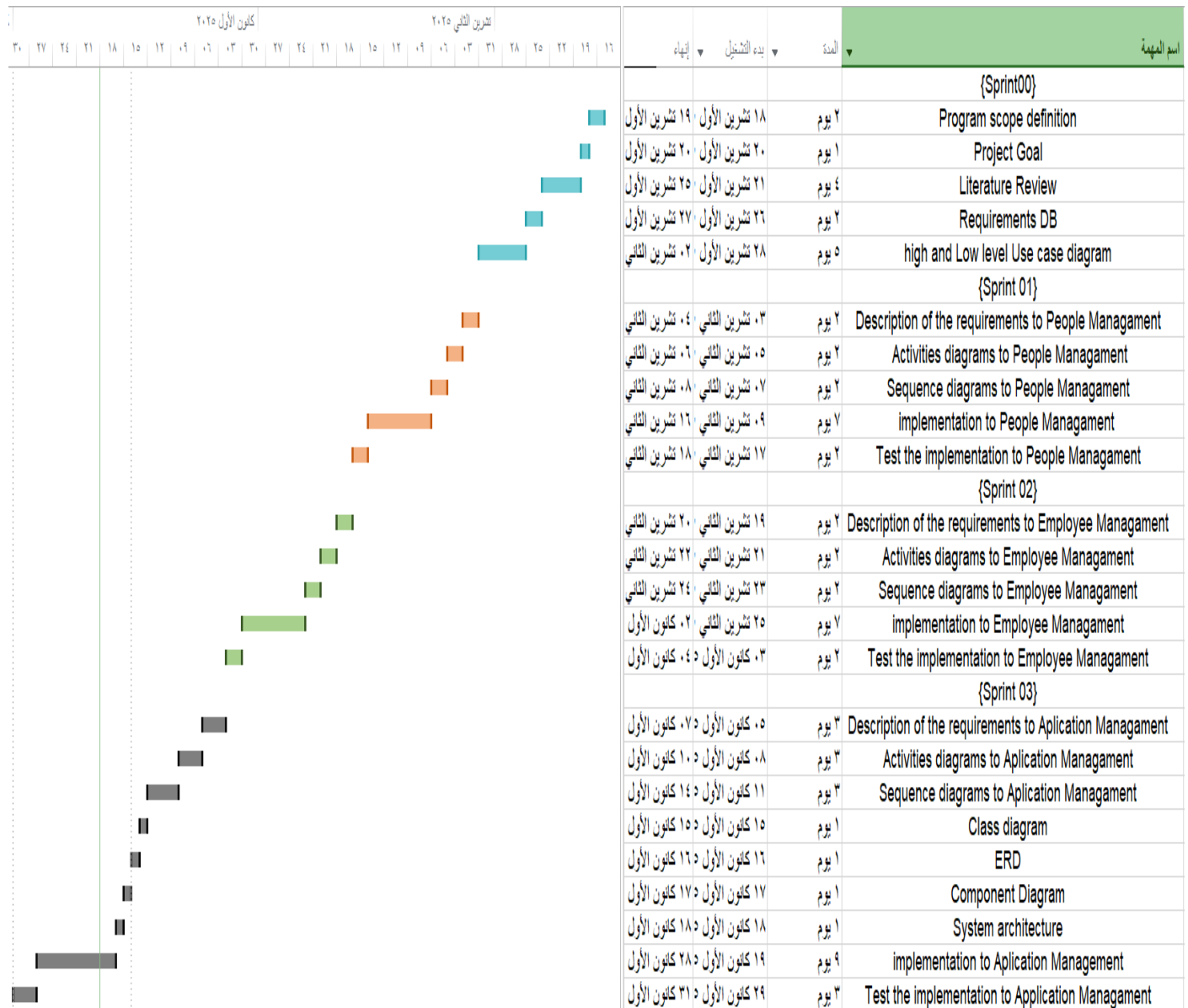
- Project Supervisor: University Supervisor
- Team Members:
 - Software Engineering Students (Development Team)

3.10 Project Schedule

Milestone	Date
Project Start Date	18/10
Requirements Analysis & SRS	25/10
System Design (UML & ERD)	10/11
Implementation Phase	15/11
Testing & Refinement	10/12
Final Submission & Delivery	31/12

4. Project plan – Gantt chart:

A document outlining tasks, deadlines, and resources needed to achieve project objectives, serving as a roadmap for successful project execution.



5. Risk Management:

the process of identifying, evaluating, and mitigating potential risks that could impact the success of a project or a team.

Risk_ID	RK-01	RK-02	RK-03	RK-04
Risk Title	Small development team	Unclear or changing requirements	Complexity of business rules	Tight project schedule
Risk Description	The project is developed by a limited number of students. If one team member becomes unavailable, the overall progress of the project may be delayed.	Misunderstanding or late clarification of system requirements (especially after changing the system to electronic submission) may lead to rework or incorrect implementation.	The driving license system contains complex rules related to requests, tests, licenses, and status transitions, which may cause implementation difficulties.	The limited timeframe between the project start and end dates may affect development, testing, and documentation quality.
Raised Date	18/10	18/10	25/10	18/10
Tracking Frequency	Weekly	Weekly	Daily	Daily
State	Active	Under Mitigation	Active	Active
Impact	High	Medium	High	Medium
Mitigation Plan	Tasks are shared and documented clearly so that each team member can assist the other when needed.	Conduct regular meetings with the supervisor and review requirements after each phase to ensure correct understanding.	Break down complex business logic into smaller modules and validate each part separately.	Follow Agile Scrum methodology with short iterations and continuous progress tracking.

6. Summary:

In conclusion, good project management is vital for successful software system development. It provides structure, ensuring projects are completed within scope and schedule. Overall, strong project management enhances the delivery of high quality software systems that meet the goals of the project.

Chapter 4 System Analysis

1. Introduction:

In this chapter we will introduce the analytical study of the system using the needed UML diagrams for system requirements modeling.

2. Requirements Elicitation:

During the requirements elicitation process, we have taken several steps to ensure a comprehensive understanding of the project. We conducted a study and analysis of similar systems, also analyzed, and anticipated the needs of the application's users.

The resulted Requirement database:

Actor	Category	Description	Req ID
All Employees	Person Management	Add a New Person	FR-1
All Employees	Person Management	Update Person Data	FR-2
All Employees	Person Management	Delete Person	FR-3
All Employees	Person Management	View Person Data	FR-4
All Employees	Person Management	Searching for a Person	FR-5
System Admin	Employee Management	Delete a Employee	FR-6
System Admin	Employee Management	Change their Password	FR-7
System Admin	Employee Management	Create a New Account for Employee	FR-8
System Admin	Employee Management	View Employee Data	FR-9
System Admin	Employee Management	Update Employee Data	FR-10
System Admin	Employee Management	Searching for a Employee	FR-11
All Employees	Employee Management	Login Accounts for Employees	FR-12
All Employees	Employee Management	View Current User Info	FR-13
All Employees	Employee Management	Change Current User's Password	FR-14
All Employees	Employee Management	Logout	FR-15
System Admin	Application Management	View Applications Type Data	FR-16
System Admin	Application Management	Edit Applications Types	FR-17
All Employees	Application Management	View Application Info	FR-18
All Employees	Application Management	Request a New Local License Application for a Person	FR-19
All Employees	Application Management	Delete Local License Application	FR-20
All Employees	Application Management	Cancel Local License Application	FR-21
All Employees	Application Management	Edit Local License Application	FR-22
All Employees	Application Management	Searching for a Request	FR-23
-	Non Functional	The System Should have User-Friendly Interface	NFR-1
-	Non Functional	The System Should Store Data Securely Using Encryption	NFR-2

3. SRS Document:

1. Introduction:

1. Purpose:

The purpose of this Software Requirements Specification (SRS) is to outline the functional and non-functional requirements for our system. This document will serve as a foundational reference to ensure alignment between stakeholders and the development team throughout the project lifecycle.

2. Project scope:

The system is designed to support the daily operations within driving centers or electronically through a single administrative user. It includes storing trainee information, organizing training schedules, recording exam results received from instructors or examiners, and issuing licenses.

3. Document overview:

The software requirements specification SRS document will be structured as follows:

1. Introduction.
2. Overall description and product features.
3. Nonfunctional requirements.
4. Conclusion.

2. Overall description and product features:

1. Product features:

The product features derived from the system's use cases are:

1- As a license officer, I want to **add a new person** so that I can register their information in the system.

Acceptance Criteria:

Fields include name, national ID, and date of birth

A success message appears after saving.

2- As a license officer, I want to **edit an existing person's information** so that I can correct or update their data.

Acceptance Criteria:

All fields except national No and ID can be edited.

A success message appears after saving changes.

Updates are stored directly in the data

3- As a license officer, I want to **view a person's information** so that I can verify their details.

4- As a license officer, I want to **delete a person** so that I can remove unnecessary records.

Acceptance Criteria:

System asks for confirmation before deletion.

A success message appears after deletion.

A person linked to applications cannot be deleted

5- As a license officer, I want to **filter people** by name or national No so that I can quickly find the required person.

Acceptance Criteria:

Search works with partial of name

Search works with ID

Results are displayed in a sorted table

6- As an admin, I want to **create an account for a new employee** so that they can access the system.

Acceptance Criteria:

Input fields: Username , password.

Username must be unique.

Success message appears after account creation.

7- As an admin, I want to **update employee information** so that I can ensure their records remain accurate and current.

Acceptance Criteria:

Editable fields: name, email, address , password.

Employee ID cannot be changed.

Success message appears after saving.

8- As an admin, I want to **view employee information** so that I can verify their details.

9- As a manager, I want to **delete an employee** so that I can ensure system security and prevent unauthorized access after they leave.

Acceptance Criteria:

System asks for confirmation before deletion.

Success message appears after deletion.

Employees with pending requests cannot be deleted.

10- As an admin, I want to **filter employees** so that I can quickly find the required employee.

Acceptance Criteria:

Results displayed in a sorted table.

Search works with partial of name

11- As an admin, I want to **change employee's password** so that I can keep their account secure.

Acceptance Criteria:

Old password must be entered for verification.

New password must meet complexity rules (min 8 chars, numbers, symbols).

Success message appears after change.

12- As a system user, I want to **log in** so that I can access the system.

Acceptance Criteria:

Login requires username and password.

Incorrect credentials show “Invalid login details”.

13- As a system user, I want to **view my account information** so that I can verify my personal details.

14- As a system user, I want to **change my password** from my account page so that I can maintain security.

Acceptance Criteria:

Old password required.

New password must meet complexity rules.

Success message appears after change.

15- As a system user, I want to **log out** so that I can end my current session.

Acceptance Criteria:

Clicking “Logout” ends the session.

Redirects to login page.

Protected pages cannot be accessed after log

16- As an admin, I want to **view application types** so that I understand available options.

Acceptance Criteria:

List of application types displayed with name and description and price.

17- As an admin, I want to **edit application types** so that I can update their details.

Acceptance Criteria:

Editable fields: name, description.

Success message appears after saving.

18- As a license officer, I want to **view application information** so that I can verify details.

19- As a license officer, I want to **submit a local driving license request for a person** so that they can obtain a local license.

Acceptance Criteria:

Success message appears after submission.

Application saved with status “New”

20- As a license officer, I want to **delete a local license application** so that I can remove unnecessary requests.

Acceptance Criteria:

System asks for confirmation before deletion.

Success message appears after deletion.

completed applications cannot be deleted

21- As a license officer, I want to **cancel a local license application** so that I can stop the process if no longer needed.

Acceptance Criteria:

System asks for confirmation before cancellation.

Success message appears after cancellation.

Application status changes to “Cancelled”.

22- As a license officer, I want to **edit a local license application** so that I can change the license class of this application.

Acceptance Criteria:

Success message appears after saving.

Changes are stored in the database.

23- As an employee, I want to **filter applications** by person’s name or status so that I can quickly find the required application.

Acceptance Criteria:

Search by status (New/Cancelled/Completed).

Results displayed in a sorted table.

2. User classes and characteristics:

Our system have two main types of users and they are:

- The System Admin:

The system admin is a high-privilege user with full access to all system functionalities. This user is responsible for managing and configuring the system, including adding, updating, and deleting system users, managing user roles and access permissions, and monitoring overall system activity. The admin ensures data integrity, oversees license issuance processes, and maintains the performance and security of the application.

- The System Operator (Employees):

The operator is the primary user of the system who interacts with the core functionalities on a daily basis. This user is responsible for managing trainee records, entering and updating personal information, scheduling training sessions, recording exam results, and submitting requests for license issuance or renewal. The operator has limited access and operates within the scope defined by the admin.

3. Assumptions:

- System Environment: The application is assumed to operate as a web-based system within official driver's license issuance centers or electronically. All workstations will be equipped with the necessary operating system (e.g., Windows) and the required software will be pre-installed.
- Database Availability: It is assumed that the database server (SQL Server or SQLite) will be available and properly configured to store and retrieve all system-related data without interruptions.
- User Technical Skills: It is assumed that system users (operators and admins) possess basic computer literacy and are trained to use the software, including navigating the interface, entering data, and generating reports.
- Access Roles and Security: It is assumed that each user will log in using their own credentials, and access will be granted based on their assigned roles (admin or operator) to ensure system security and data integrity.

3. *Nonfunctional requirements:*

1. **RE-NF-01(Security)**: Access permissions must be controlled based on roles (Role-Based Access Control).
2. **RE-NF-02(Maintainability)**: The system architecture should support modifying or updating specific modules without impacting the entire system. Comprehensive documentation must be provided for all software components to facilitate future maintenance.
3. **RE-NF-03(Usability)**: The user interface should be intuitive and easy to use for all user groups. Full support for the Arabic language must be implemented across all interfaces and reports.

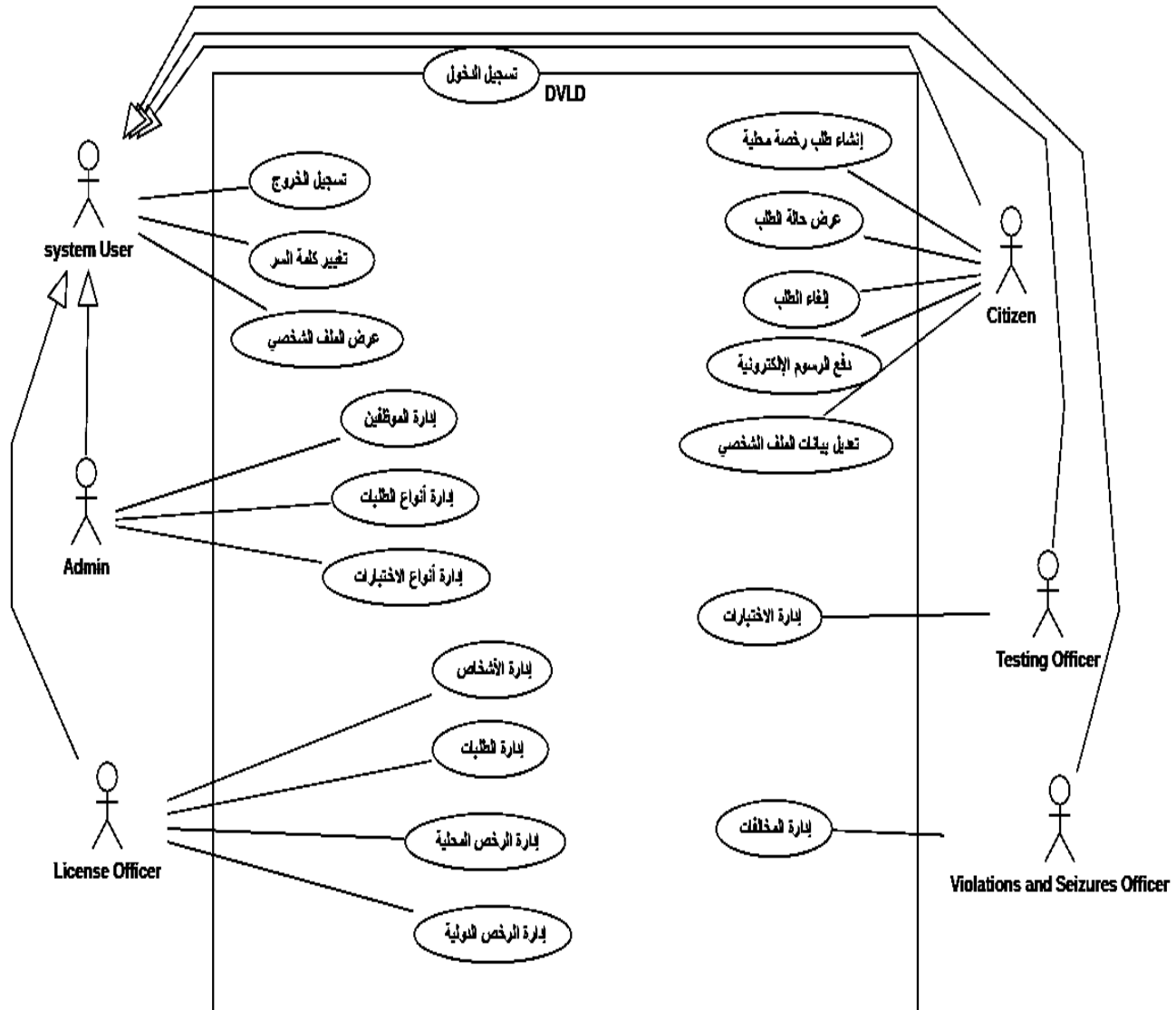
4. *Conclusion:*

This Software Requirements Specification (SRS) document outlines the essential functional and non-functional requirements, user roles, system features, and underlying assumptions necessary for the successful development of the Driving License Management System. By adhering to these specifications, the system will provide a reliable and user-friendly solution for managing trainee records, scheduling exams, issuing licenses, and ensuring secure administrative control.

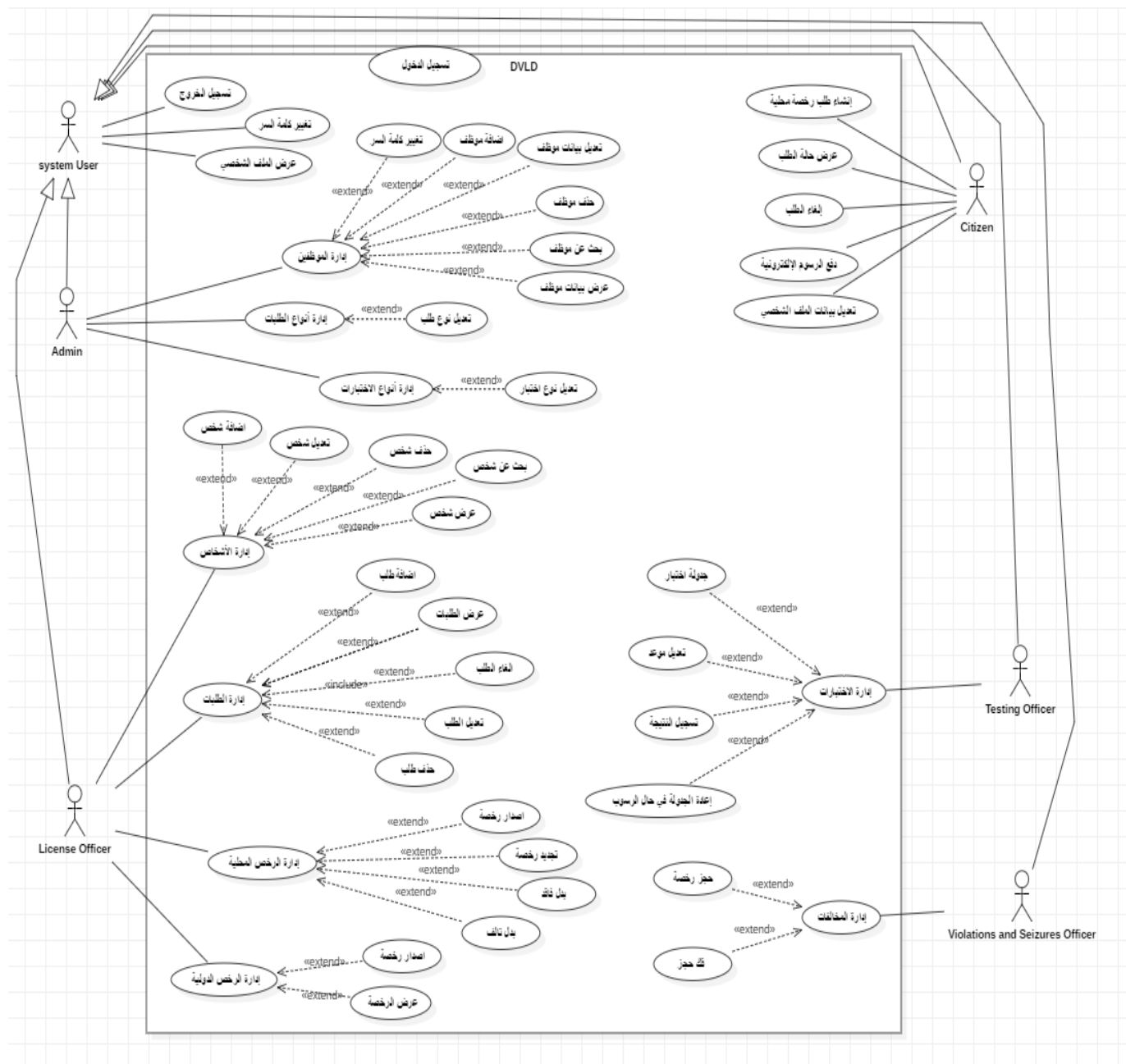
The clear definition of user roles, interface expectations, and performance standards will help guide the development process efficiently, ensuring that the final product meets both organizational goals and end-user needs. Ultimately, this system aims to streamline operations in driving license centers, reduce human error, and enhance service quality through digital transformation.

4. Requirements modeling:

High-level use case diagram: High-level use case diagrams give us an overview of system behavior and help to superficially define system requirements.



Low-level use case diagram: Low-level use case diagrams show us the system's behavior and help to define system requirements in detail.



System features – use cases:

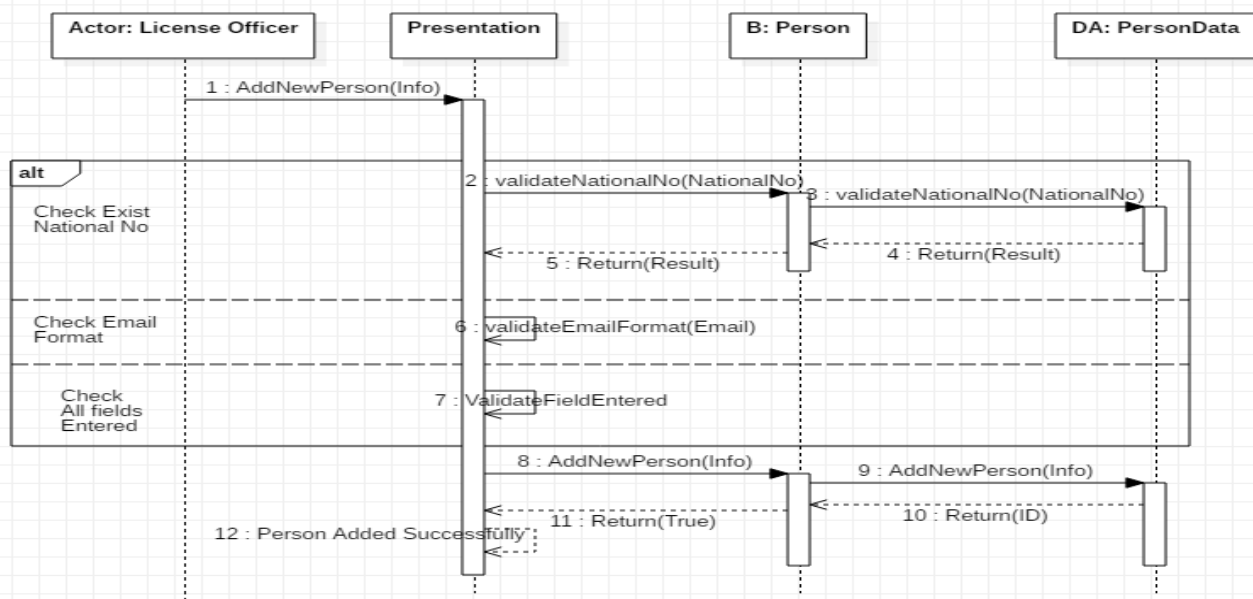
Description of the requirement to login:

Use Case Title	Login
Actors	System User
Preconditions	None
Main Flow	• The user clicks the Login button. • The system displays the login interface. • The user enters the username and password and clicks the Login button. • The system verifies the validity of the username and password and checks the account status. • The system displays the main interface to the user.
Alternative Flow	• If the entered username or password is incorrect, the system displays an error message and allows the user to re-enter the credentials. • If the username and password are correct but the account is inactive, the system displays an error message and allows the user to re-enter the credentials.
Postcondition	None

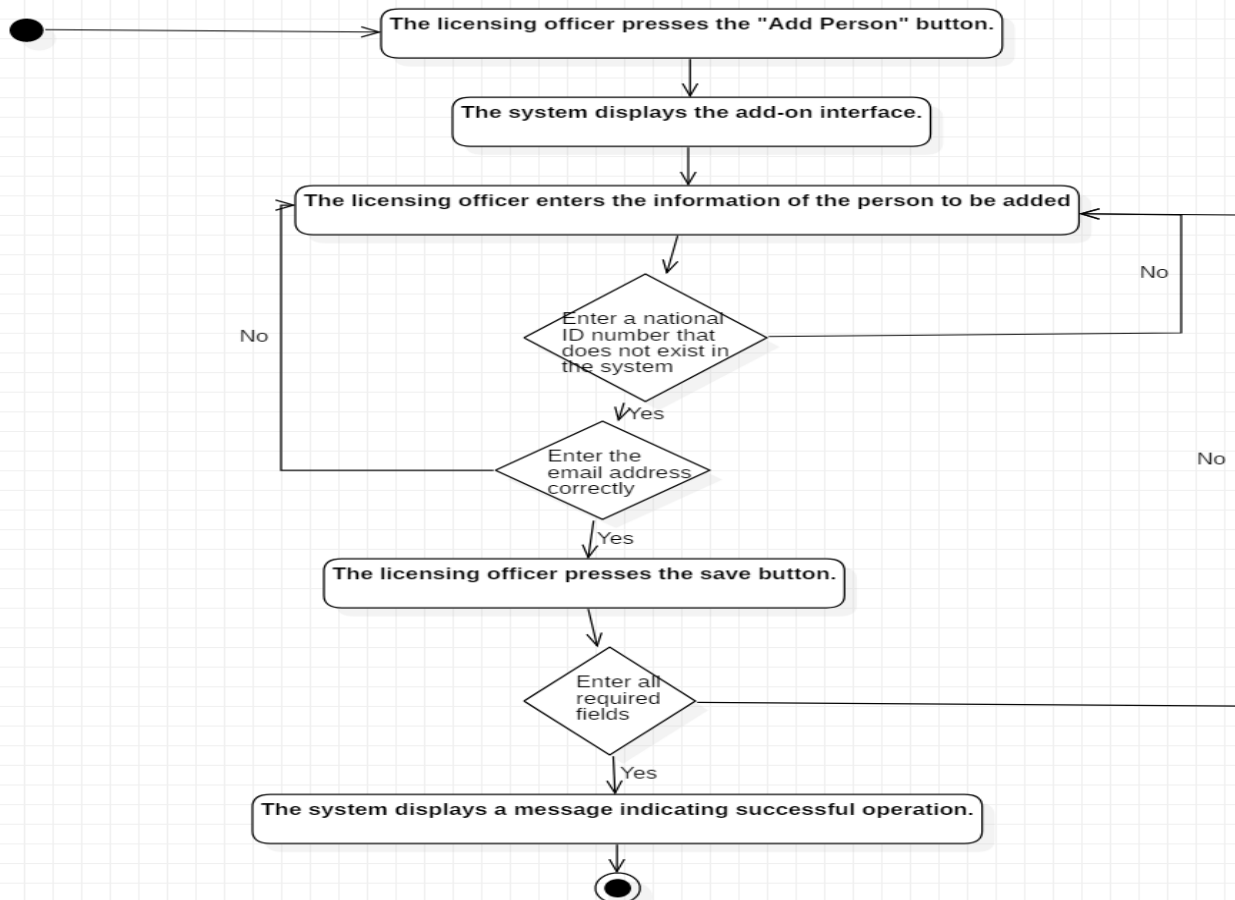
Description of the requirement to add person:

Use Case Title	Add Person
Actors	License Officer
Preconditions	Login
Main Flow	1. The License Officer clicks on the People icon. 2. The system displays an interface with the list of existing people. 3. The License Officer clicks the Add Person button. 4. The system displays the "Add Person" interface. 5. The License Officer enters the information of the person to be added. 6. The License Officer clicks the Save button. 7. The system displays a success message confirming the addition.
Alternative Flow	During Step 5: 1. If the License Officer enters a National ID that already exists, the system displays an error alert and requires it to be changed. 2. If the License Officer enters an Email in an incorrect format, the system displays an error alert and requires a valid format. During Step 6: 3. If any mandatory field is left empty, the system displays an error alert and forces the License Officer to fill in all required fields.
Postcondition	None

Sequence diagram Add Person:



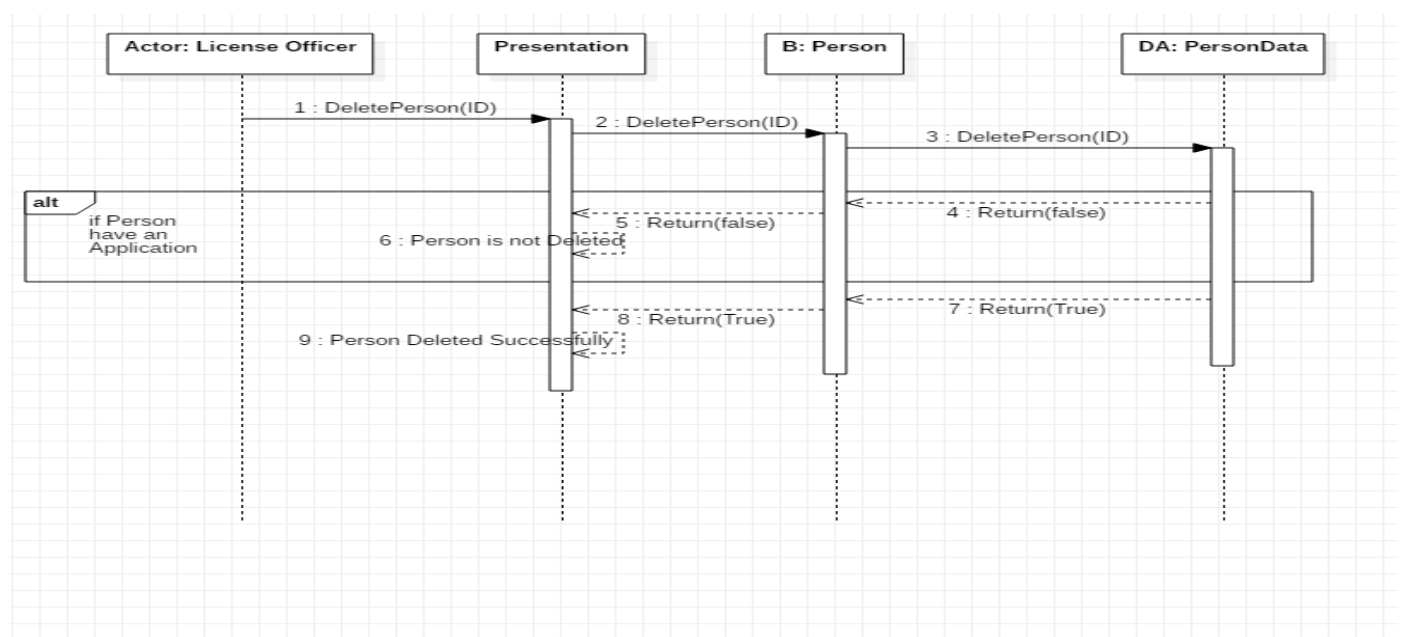
Activity Diagram Add New Person:



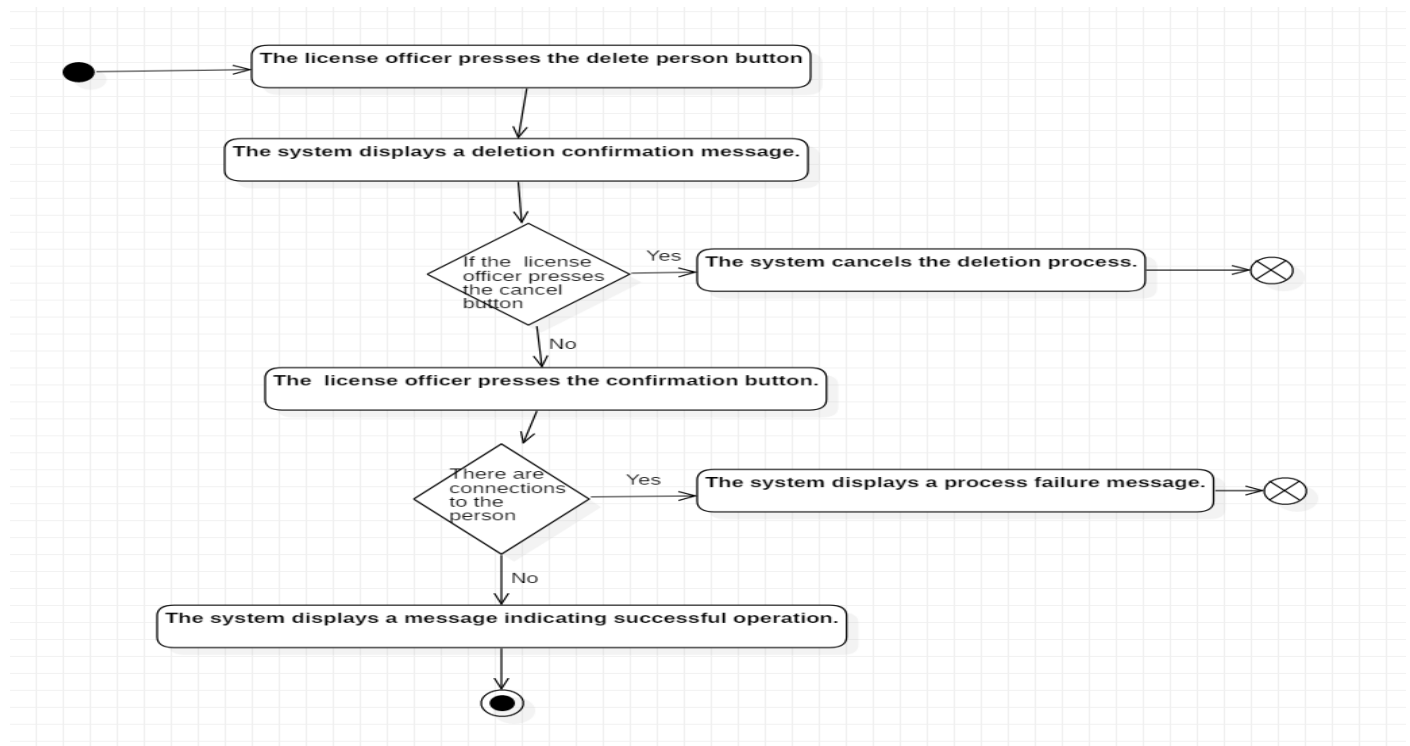
Description of the requirement to Delete Person:

Use Case Title	Delete Person
Actors	License Officer
Preconditions	Login
Main Flow	1. The License Officer clicks on the People icon. 2. The system displays an interface with the list of existing people. 3. The License Officer selects the target person and clicks the Delete Person button. 4. The system displays a confirmation message for the deletion. 5. The License Officer clicks the Confirm button. 6. The system displays a success message confirming the deletion.
Alternative Flow	At Step 4: 1. If the License Officer clicks Cancel, the system aborts the deletion process. At Step 5: 2. If there is data linked/associated with the person to be deleted, the system displays a message preventing the deletion due to existing related records.
Postcondition	None

Sequence diagram Delete Person:



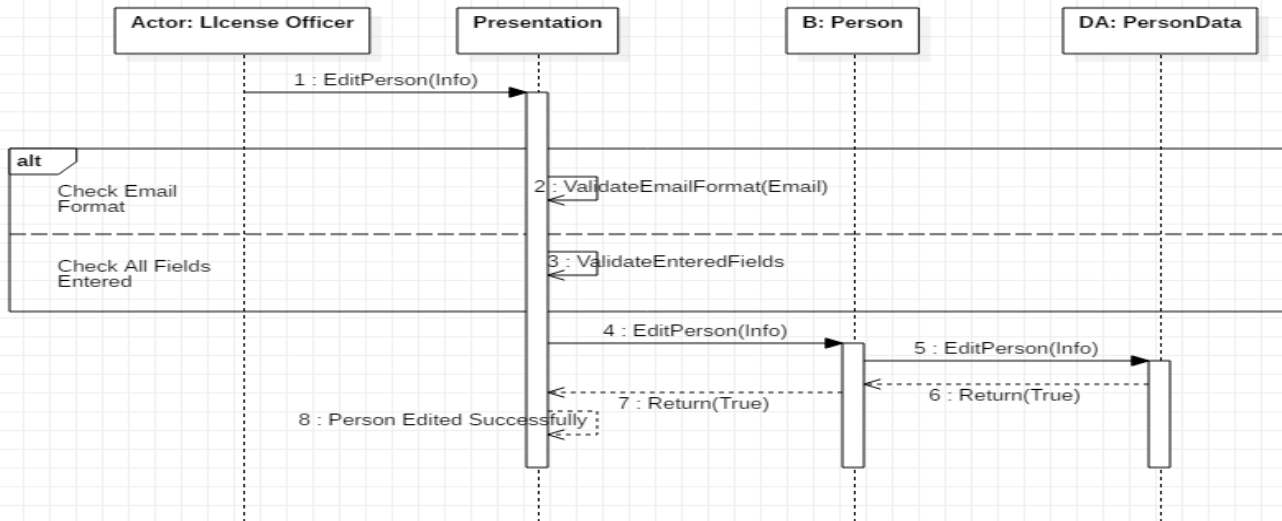
Activity diagram Delete Person:



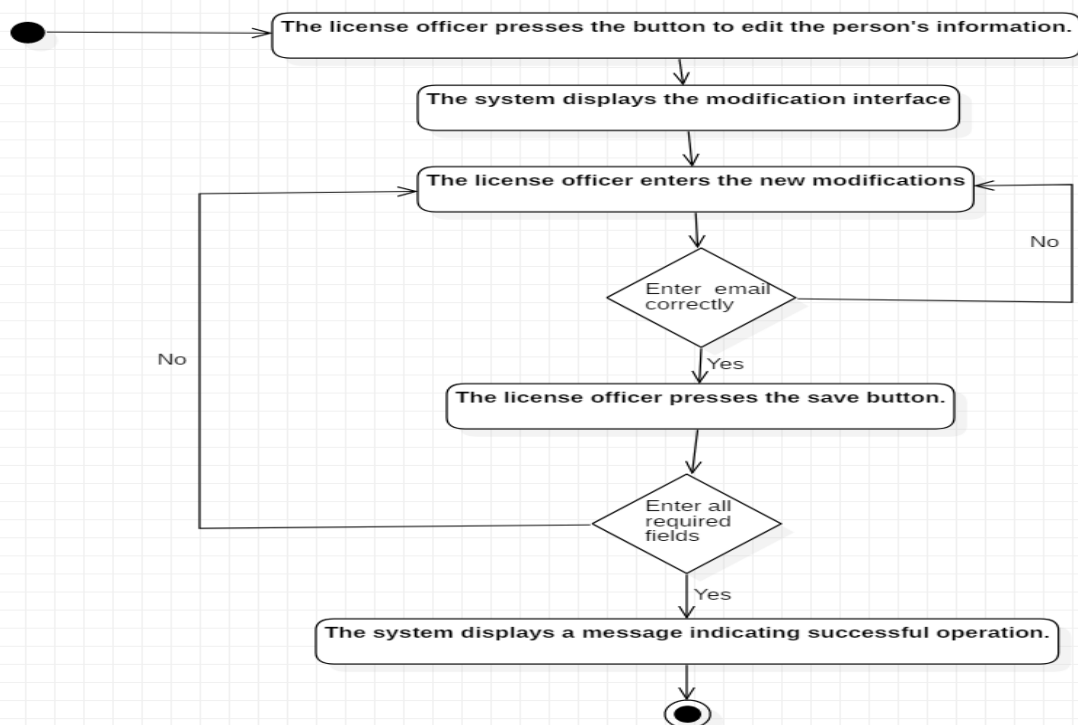
Description of the requirement to Update Person:

Use Case Title	Update Person Information
Actors	License Officer
Preconditions	Login
Main Flow	1. The License Officer clicks on the People icon. 2. The system displays an interface with the list of existing people. 3. The License Officer selects the target person and clicks the Update Person Information button. 4. The system displays the Edit interface. 5. The License Officer enters the new information. 6. The License Officer clicks the Save button. 7. The system displays a success message confirming the update process.
Alternative Flow	During Step 5: 1. If the License Officer enters an Email in an incorrect format, the system displays an error alert and requires a valid format. During Step 6: 2. If any mandatory field is left empty, the system displays an error alert and forces the user to fill in all required fields.
Postcondition	None

Sequence diagram Update Person:



Activity Diagram Update Person:



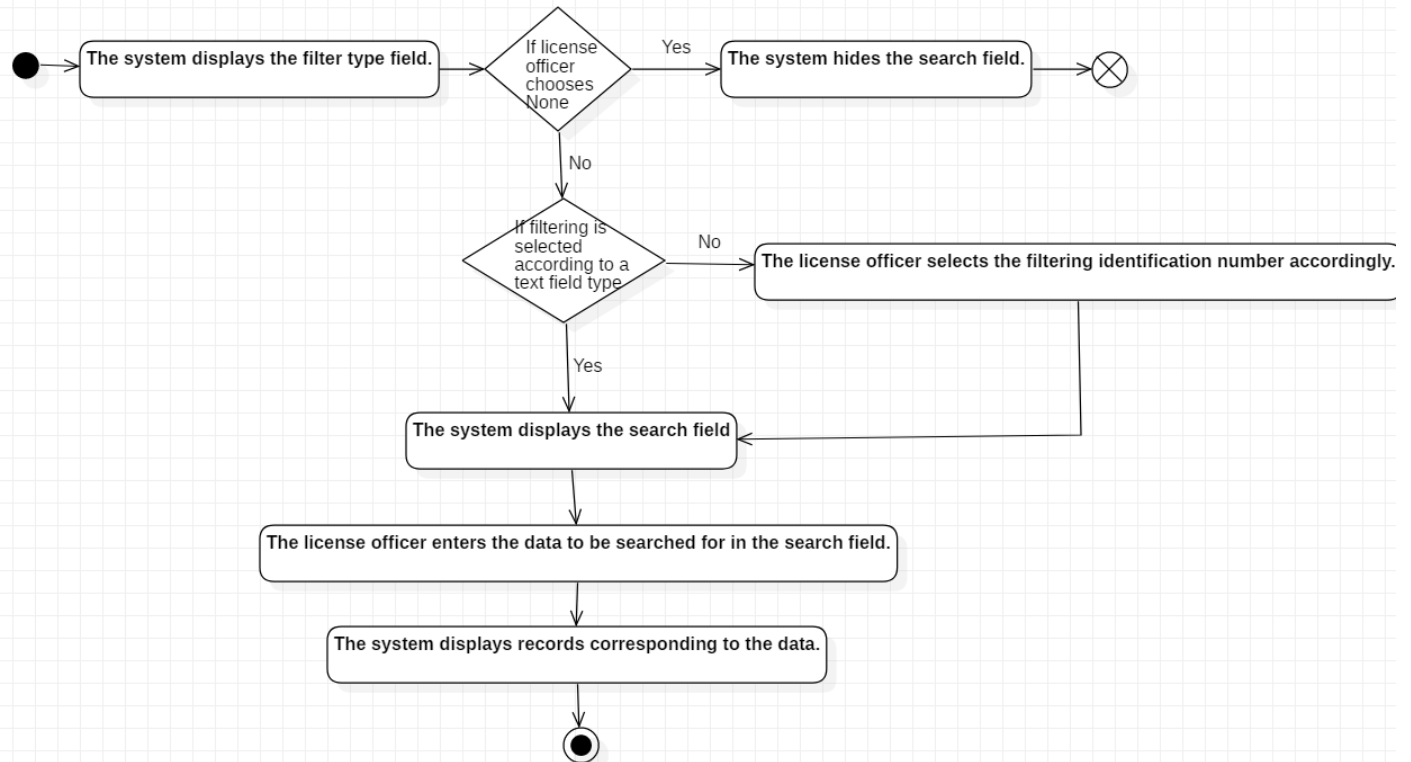
Description of the requirement to Filter People :

Use Case Title	Filter People
Actors	License Officer
Preconditions	Login
Main Flow	1. The License Officer clicks on the People icon. 2. The system displays the interface with the existing people and shows a Filter Type selection field. 3. The License Officer selects ID from the list of options. 4. The system displays a text input field. 5. The License Officer starts typing the ID they wish to search for in the field. 6. The system displays the record matching the entered ID.
Alternative Flow	1. If the License Officer selects "None", the system hides the search field and displays all existing records. 2. If the License Officer selects a text-based filter type, the system displays the records matching the entered data in real-time while typing
Postcondition	None

Sequence diagram Filter People:



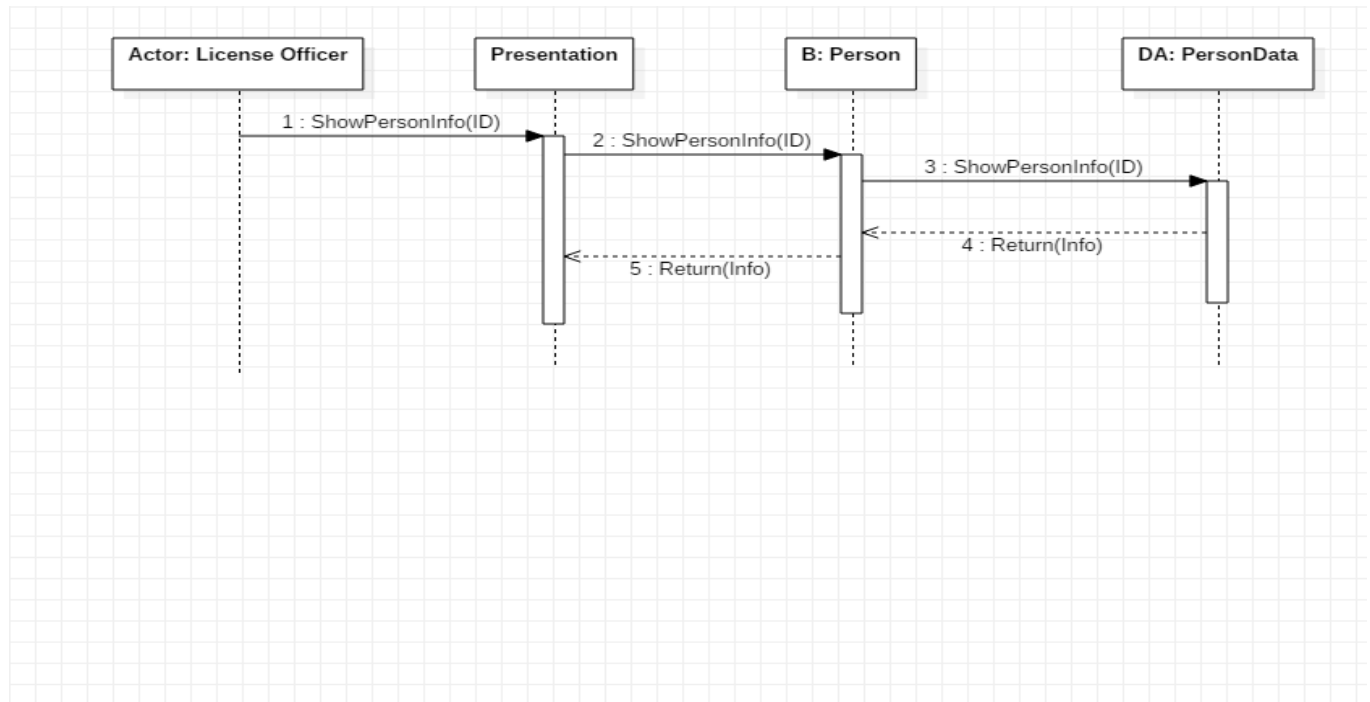
Activity Diagram Filter People:



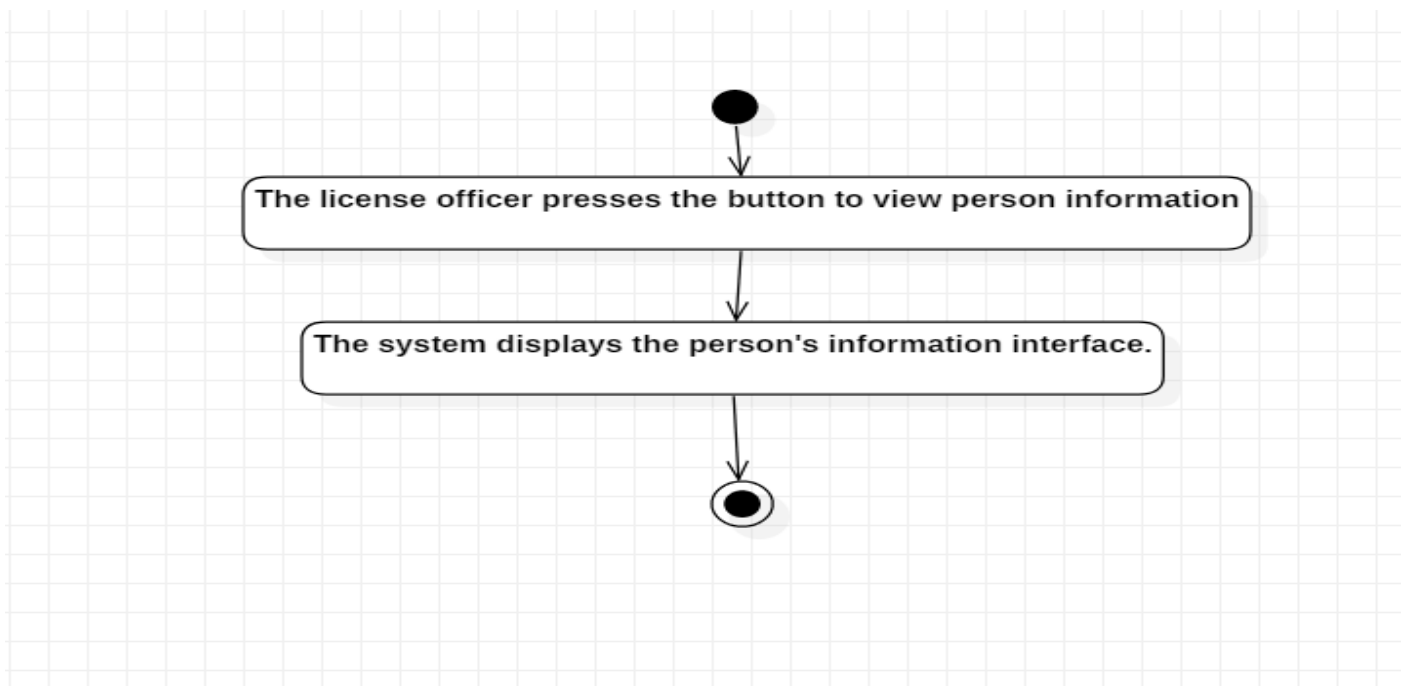
Description of the requirement to Show Person Information:

Use Case Title	View Person Information
Actors	License Officer
Preconditions	Login
Main Flow	1. The License Officer clicks on the People icon. 2. The system displays an interface with the list of existing people. 3. The License Officer selects the target person and clicks the View Person Information button. 4. The system displays the interface containing the person's specific information.
Alternative Flow	None
Postcondition	None

Sequence diagram Show Person Information:



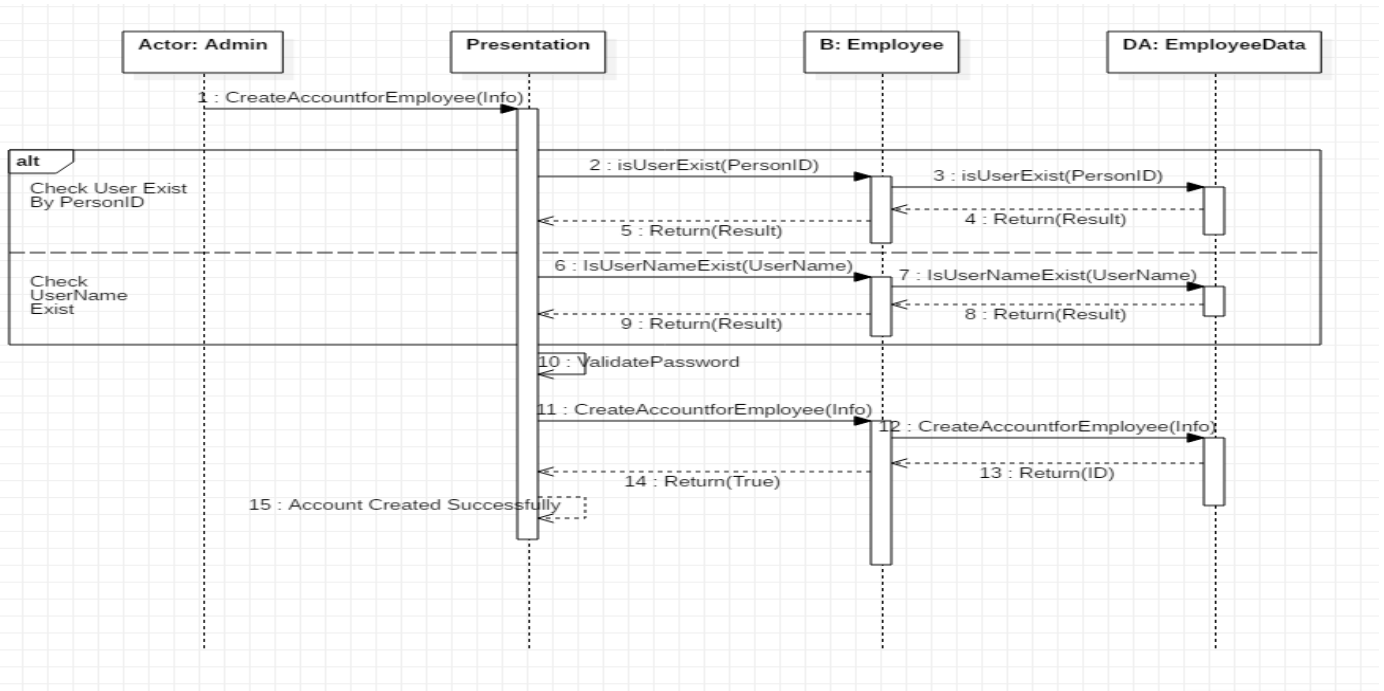
Activity Diagram Show Person Information :



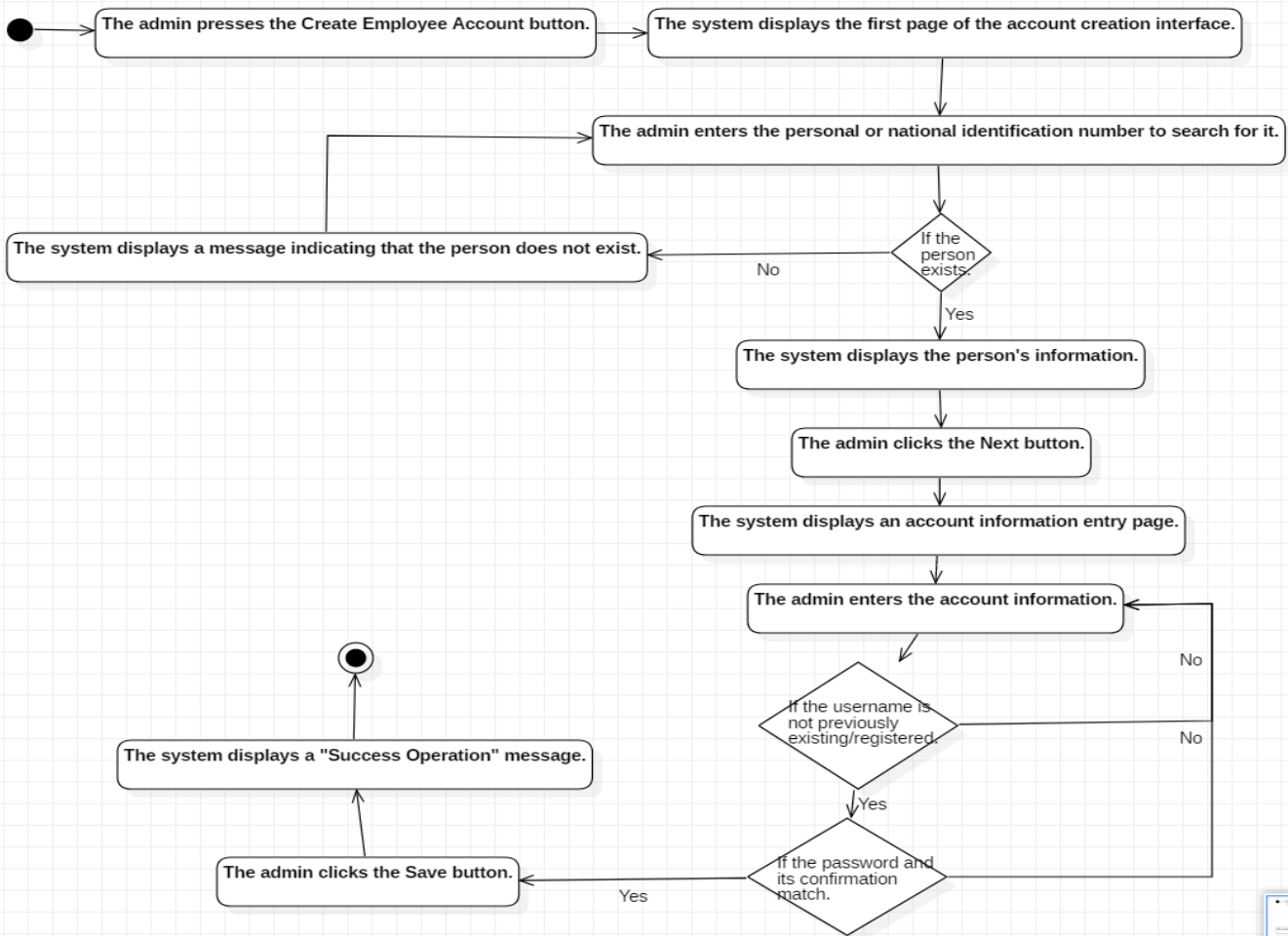
Description of the requirement to Create Account For Employee :

Use Case Title	Create Employee Account
Actors	Admin
Preconditions	Login
Main Flow	1. The Admin clicks on the Employees icon. 2. The system displays an interface with the list of existing employees. 3. The Admin clicks the Create Employee Account button. 4. The system displays the first page of the account creation interface. 5. The Admin searches for the person for whom they want to create an account. 6. The system displays the person's information. 7. The Admin clicks the Next button. 8. The system displays the account information entry page. 9. The Admin enters the account details and clicks the Save button. 10. The system displays a success message confirming the operation
Alternative Flow	At Step 6: 1. If the person is not found, the system displays an error message. At Step 9: 2. If the Admin enters a Username that already exists, the system displays an error notification: "Username already exists." 3. If the Admin enters a Password Confirmation that does not match the password, the system displays an error notification and forces the Admin to match them.
Postcondition	None

Sequence diagram Create Account For Employee:



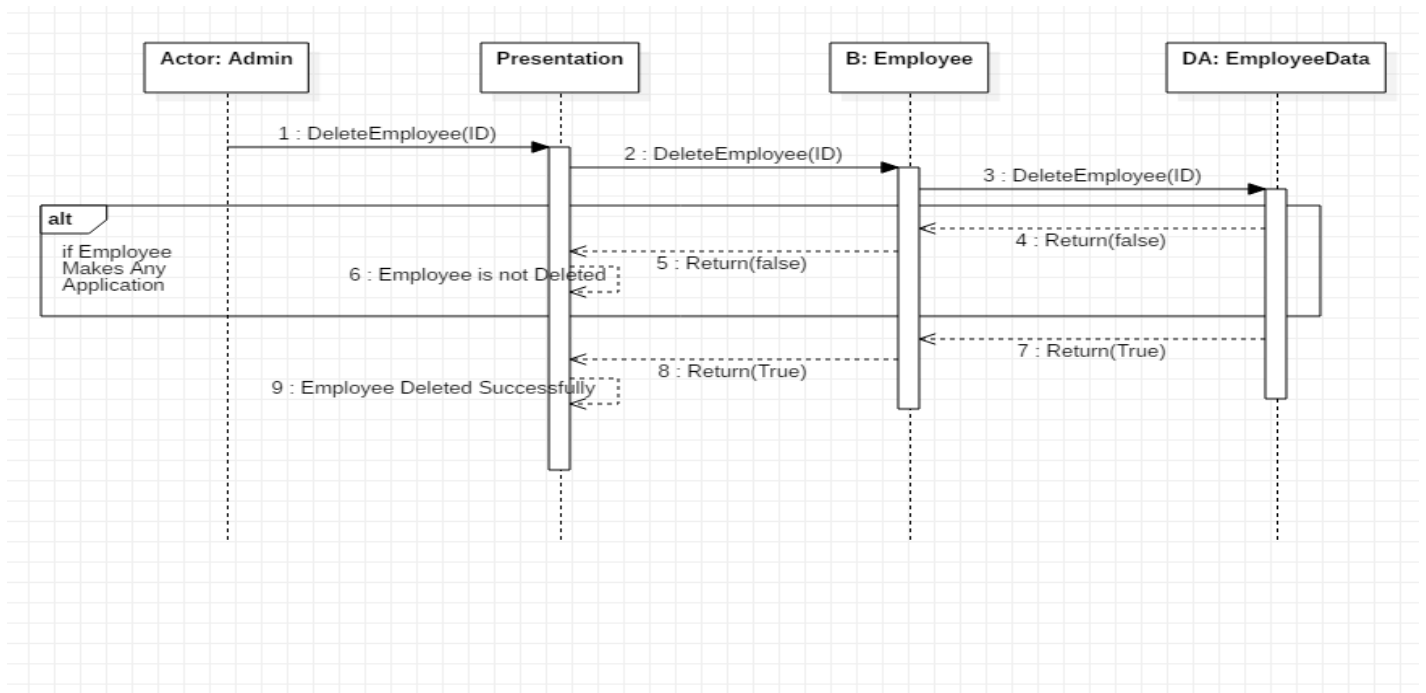
Activity Diagram Create Account For Employee:



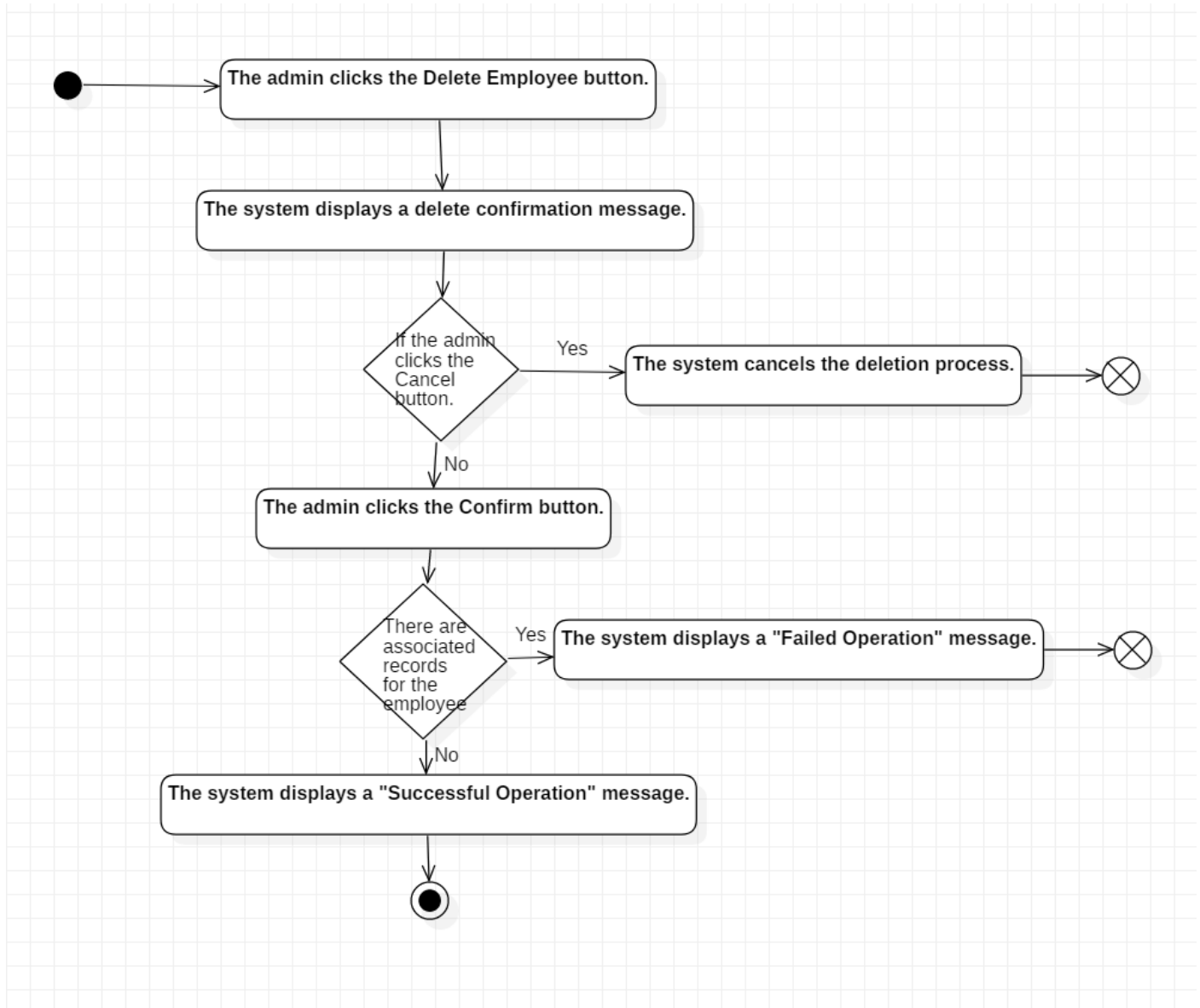
Description of the requirement to Delete Employee:

Use Case Title	Delete Employee
Actors	Admin
Preconditions	Login
Main Flow	1. The Admin clicks on the Employees icon. 2. The system displays an interface with the list of existing employees. 3. The Admin selects the target employee and clicks the Delete Employee button. 4. The system displays a confirmation message. 5. The Admin clicks the Confirm button. 6. The system displays a success message confirming the operation.
Alternative Flow	At Step 5: 1. If the Admin clicks Cancel, the system aborts the deletion process. At Step 6: 2. If the employee to be deleted has performed operations within the system, the system displays an operation failure message (due to existing activity logs).
Postcondition	None

Sequence diagram Delete Employee:



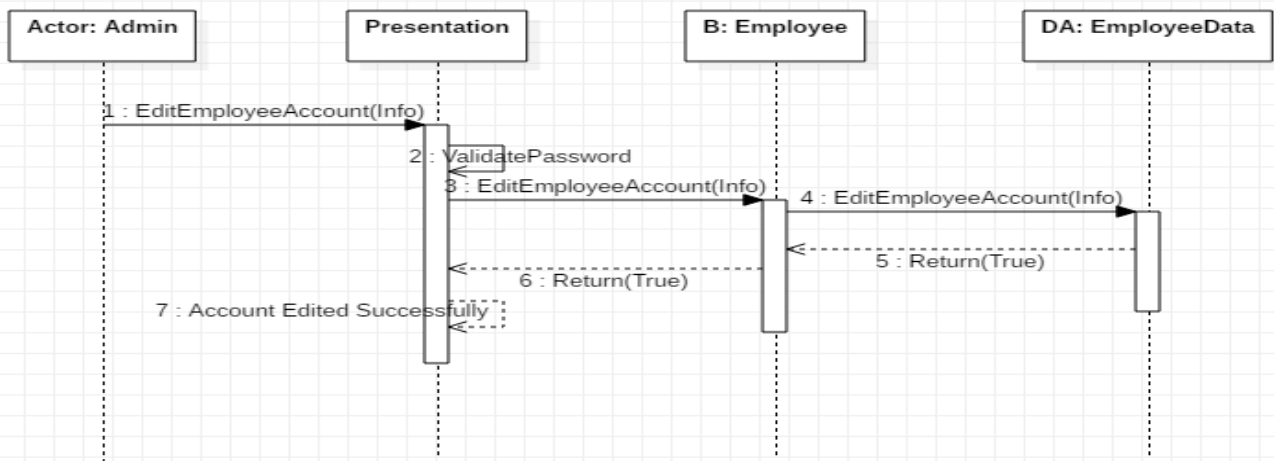
Activity Diagram Delete Employee:



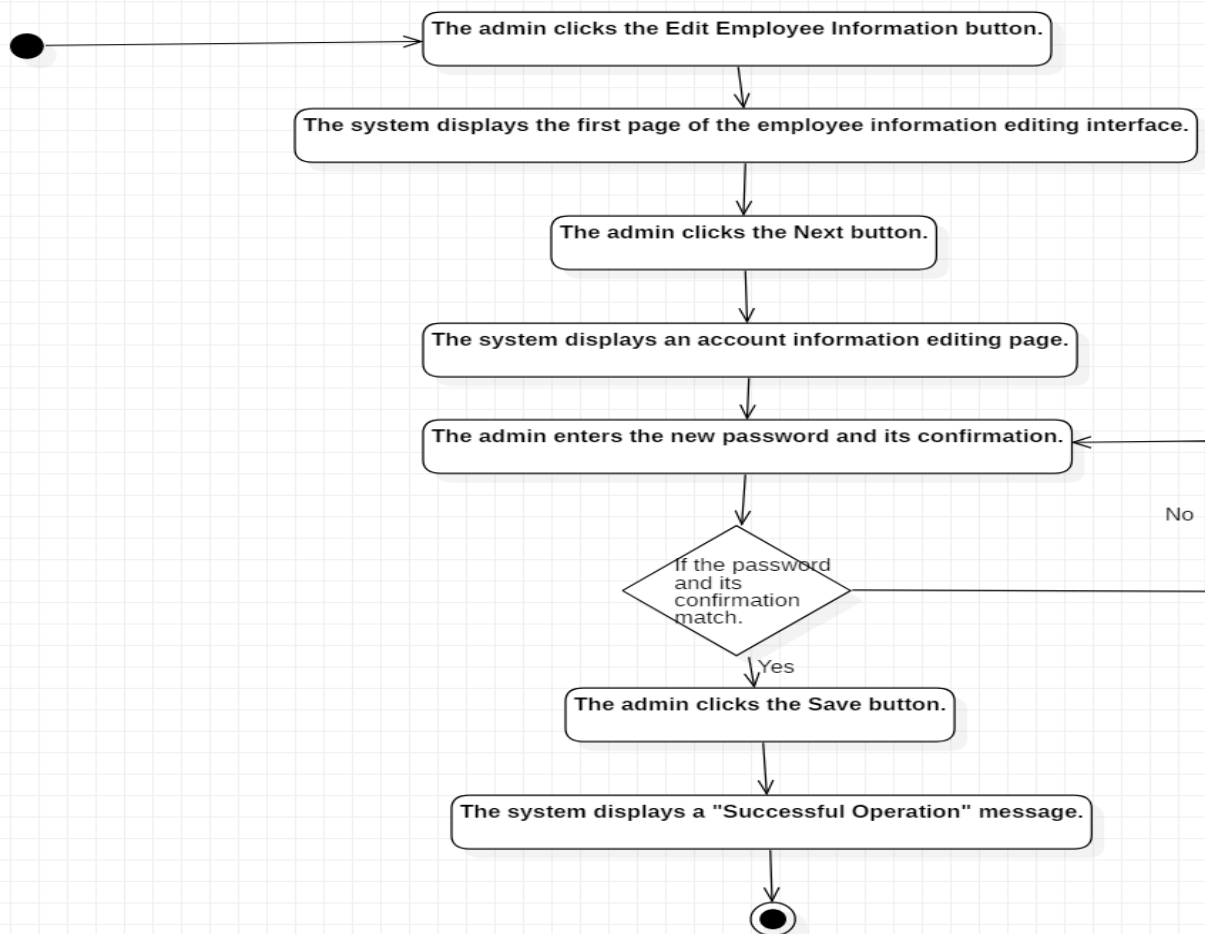
Description of the requirement to Update Employee:

Use Case Title	Update Employee Information
Actors	Admin
Preconditions	Login
Main Flow	1. The Admin clicks on the Employees icon. 2. The system displays an interface with the list of existing employees. 3. The Admin selects the target employee and clicks the Update Employee Information button. 4. The system displays the first page of the employee information update interface. 5. The Admin clicks the Next button. 6. The system displays the account information page. 7. The Admin enters the new password and confirms the password. 8. The Admin clicks the Save button. 9. The system displays a success message confirming the operation.
Alternative Flow	At Step 7: 1. If the Admin enters a Password Confirmation that does not match the password, the system displays an error alert and forces the Admin to match them.
Postcondition	None

Sequence diagram Update Employee:



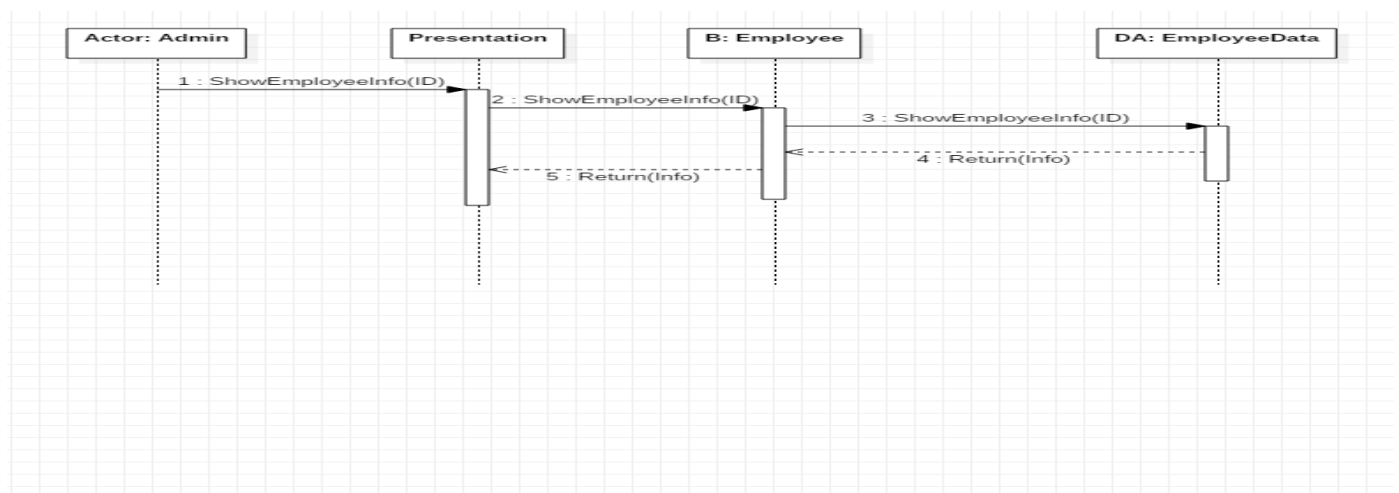
Activity Diagram Update Employee:



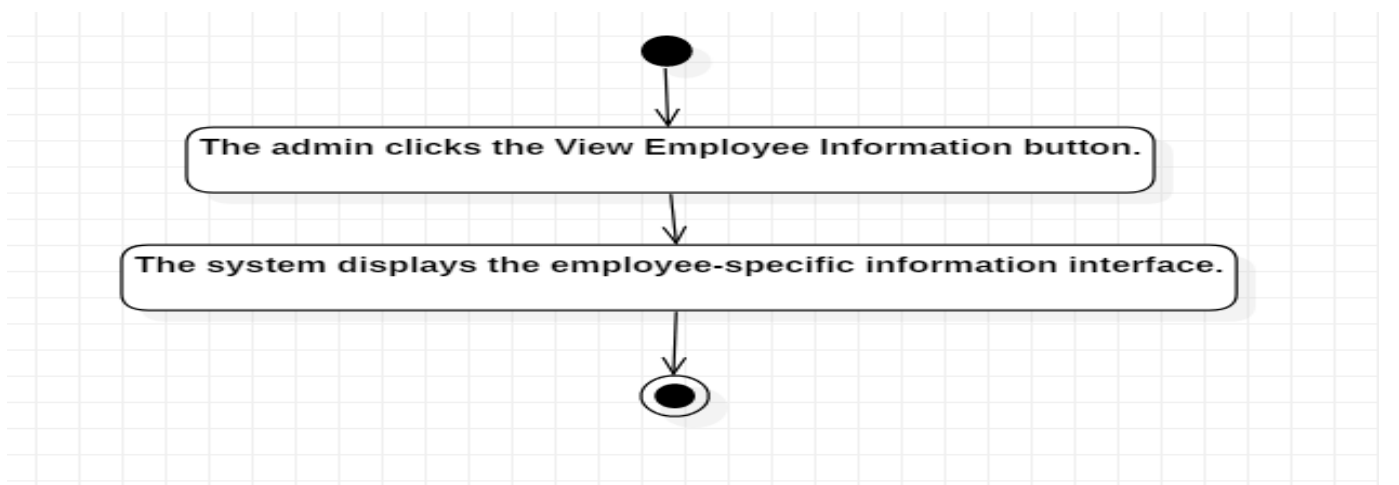
Description of the requirement to Show Employee's Information:

Use Case Title	View Employee Information
Actors	Admin
Preconditions	Login
Main Flow	1. The Admin clicks on the Employees icon. 2. The system displays an interface with the list of existing employees. 3. The Admin selects the target employee and clicks the View Employee Information button. 4. The system displays the employee information interface.
Alternative Flow	None
Postcondition	None

Sequence diagram Show Employee's Information:



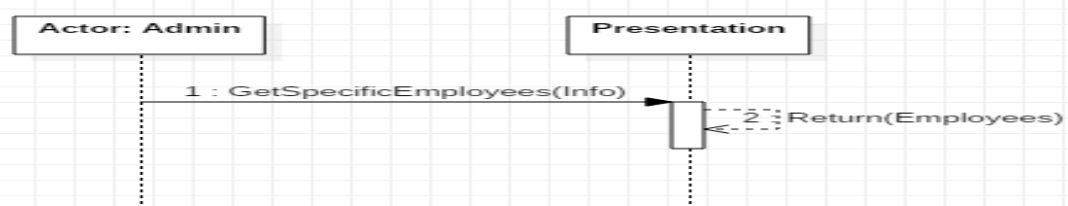
Activity Diagram Show Employee's Information :



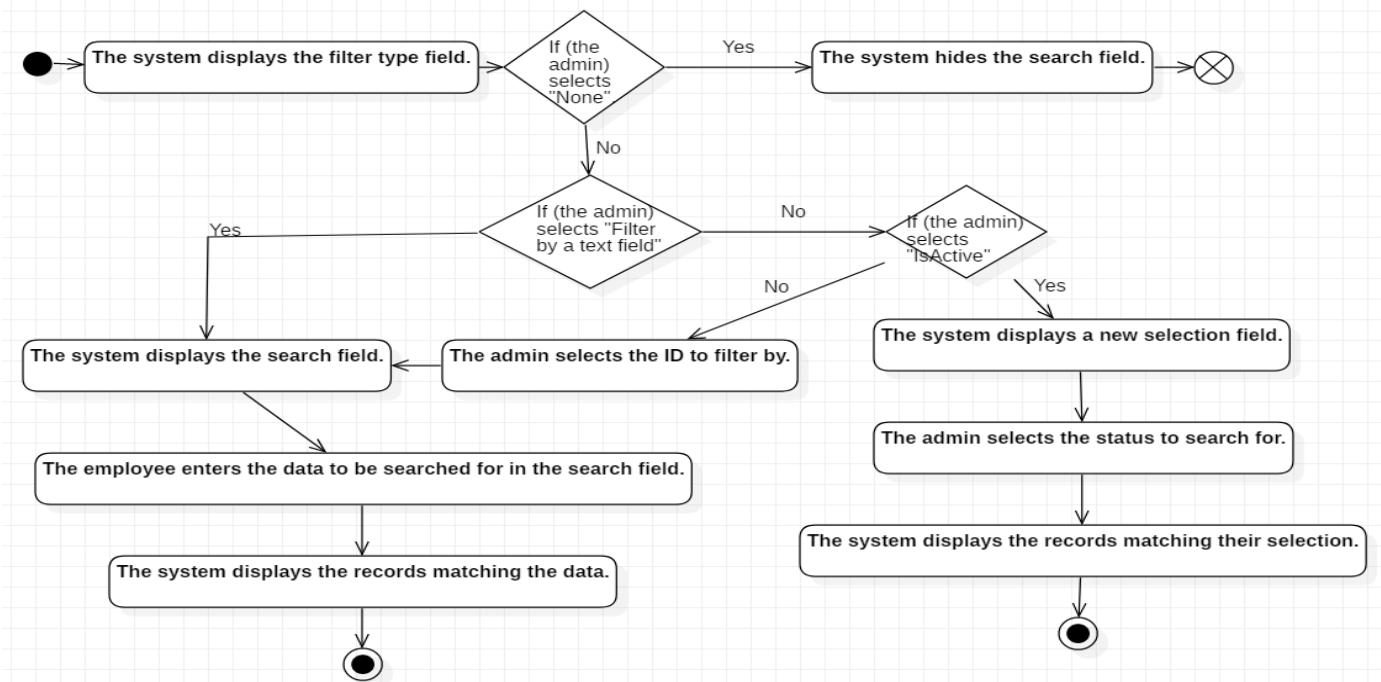
Description of the requirement to Filter Employees:

Use Case Title	Filter Employees
Actors	Admin
Preconditions	Login
Main Flow	1. The Admin clicks on the Employees icon. 2. The system displays an interface with the list of existing employees and a Filter Type selection field. 3. The Admin selects ID from the list of options. 4. The system displays a text input field. 5. The Admin starts typing the ID they wish to search for in the field. 6. The system displays the record matching the entered ID.
Alternative Flow	1. If the Admin selects "None", the system hides the search field and displays all existing records. 2. If the Admin selects a text-based filter type, the system displays the records matching the entered data in real-time while typing. 3. If the Admin selects the "Is Active" filter type, the system displays a new selection field (Dropdown) instead of the text field: - I - If the Admin selects "Yes", the system displays all active records. - II - If the Admin selects "No", the system displays all inactive records.
Postcondition	None

Sequence diagram Filter Employees:



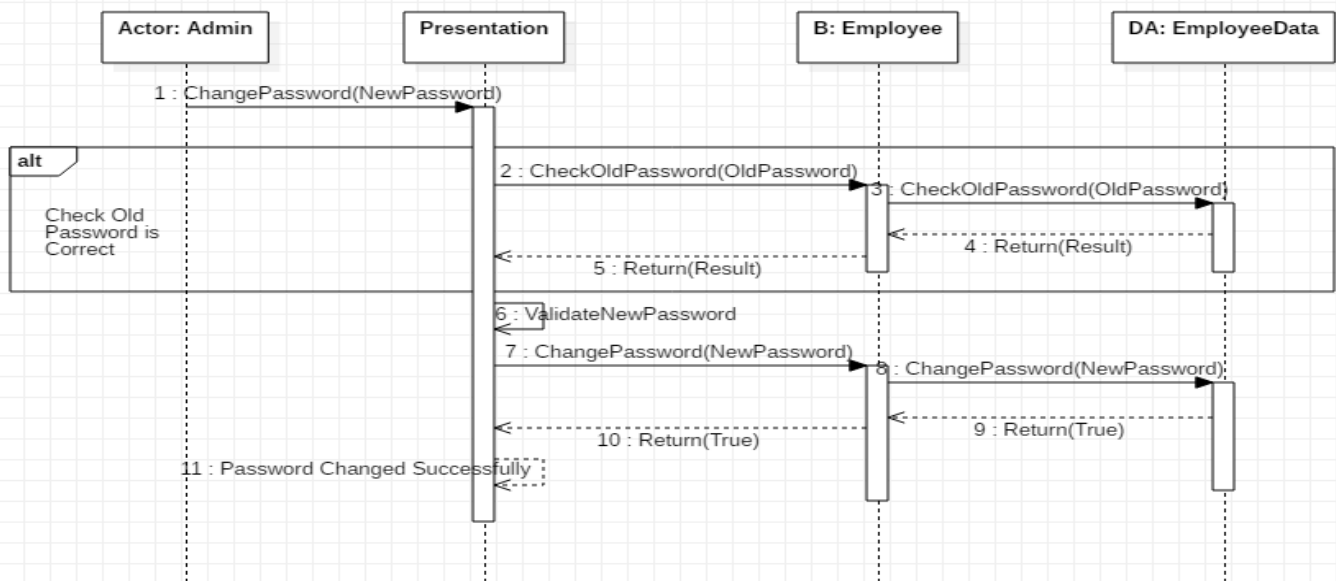
Activity diagram Filter Employees:



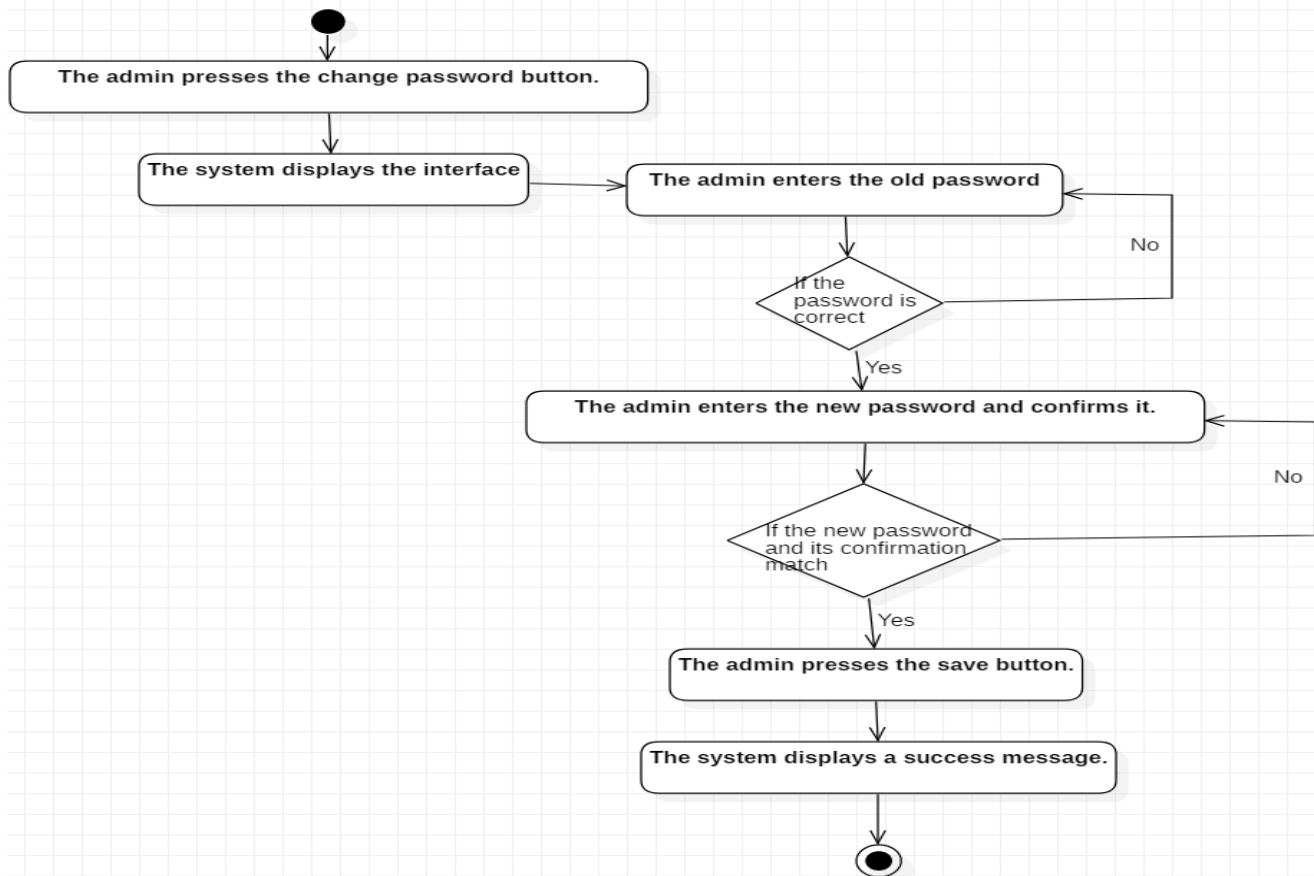
Description of the requirement to Change Password:

Use Case Title	Change Employee Password
Actors	Admin
Preconditions	Login
Main Flow	1. The Admin clicks on the Employees icon. 2. The system displays an interface with the list of existing employees. 3. The Admin selects the target employee and clicks the Change Employee Password button. 4. The system displays the Change Employee Password interface. 5. The Admin enters the Current (Old) Password. 6. The Admin enters the New Password and the Password Confirmation. 7. The Admin clicks the Save button. 8. The system displays a success message confirming the operation.
Alternative Flow	At Step 5: 1. If the Admin enters an incorrect Old Password, the system displays an error notification and requires the correct password to be entered. At Step 6: 2. If the Admin enters a Password Confirmation that does not match the new password, the system displays an error notification and forces the Admin to match them.
Postcondition	None

Sequence diagram Change Password:



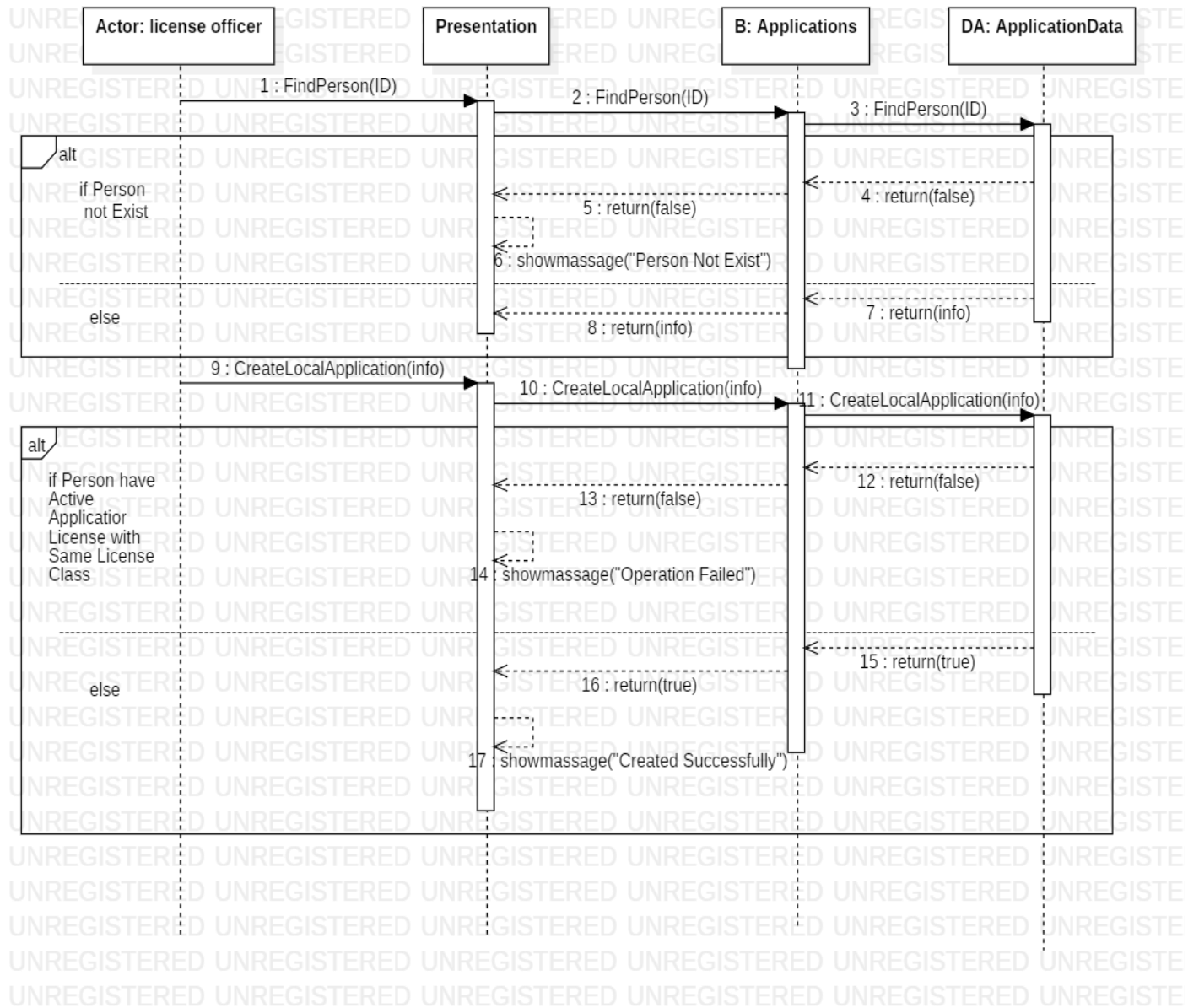
Activity Diagram Change Password :



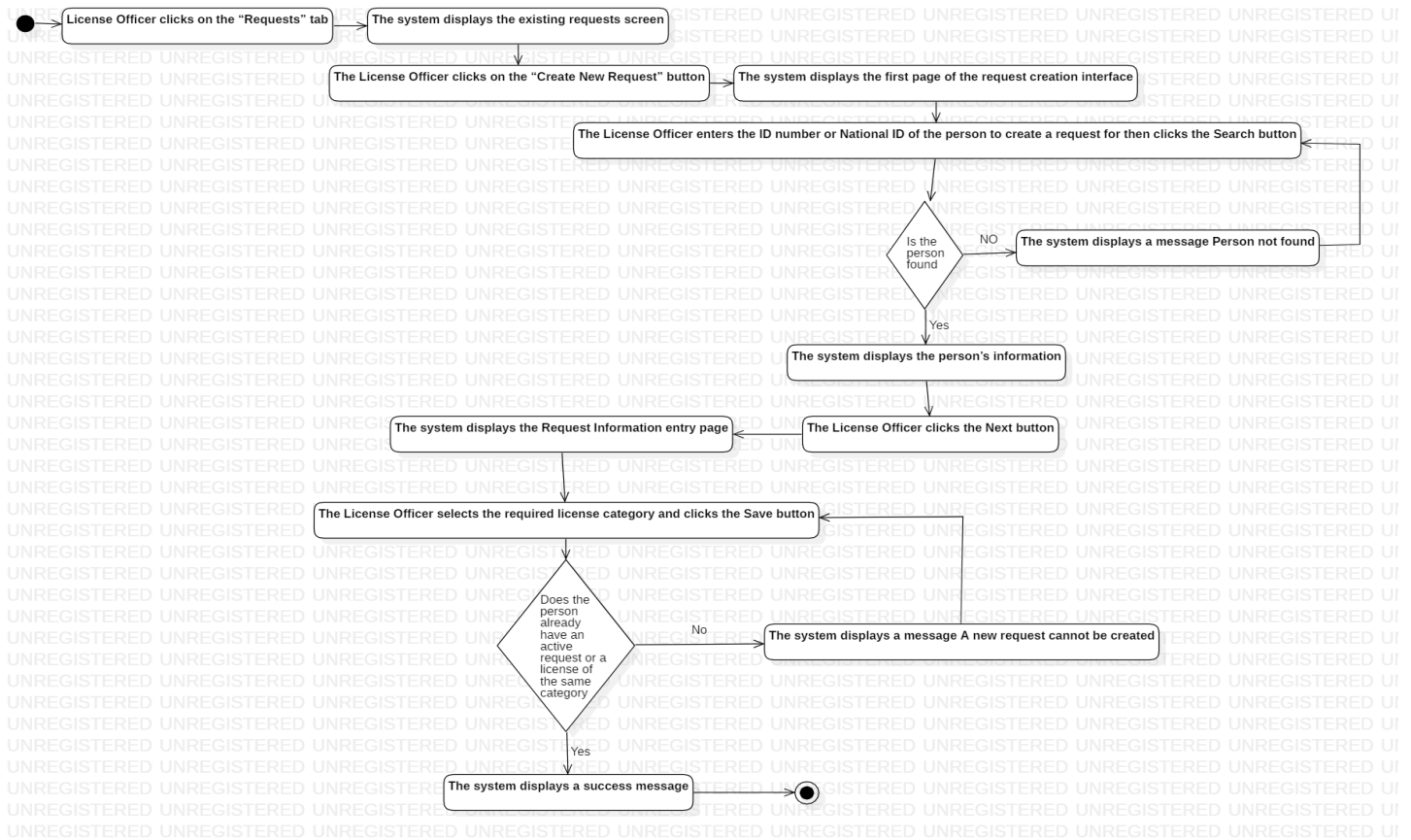
Description of the requirement to Create Local License:

Use Case Title	Create Local License Application
Actors	License Officer
Preconditions	Login
Main Flow	1. The License Officer clicks on the Applications icon. 2. The system displays an interface with the existing applications. 3. The License Officer clicks the Create New Application button. 4. The system displays the first page of the application creation interface. 5. The License Officer enters the ID or National No. of the person and clicks the Search button. 6. The system displays the person's information. 7. The License Officer clicks the Next button. 8. The system displays the application information entry page. 9. The License Officer selects the License Class to be requested and clicks the Save button. 10. The system displays a success message confirming the operation.
Alternative Flow	At Step 6: 1. If the person is not found, the system displays an error message. At Step 9: 2. If the person already has an active application of the same type, the system displays an error message. 3. If the person already holds a license of the same class, the system displays an error message.
Postcondition	None

Sequence diagram Create Local License:



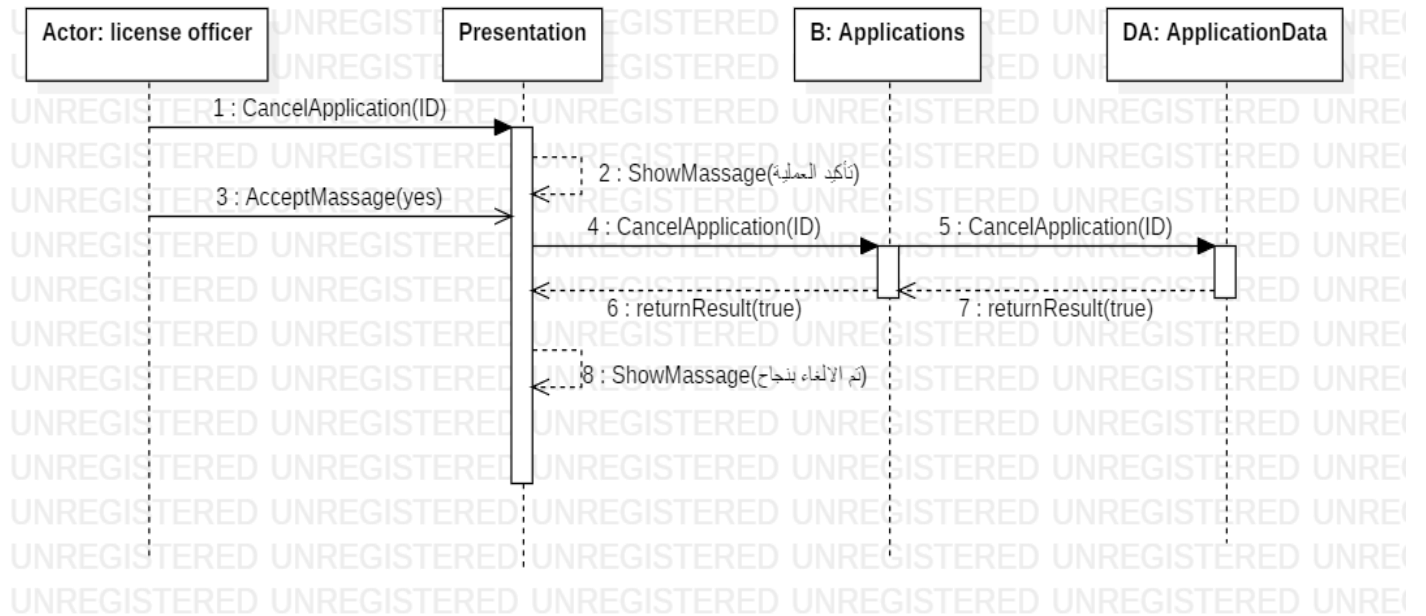
Activity Diagram Create Local License :



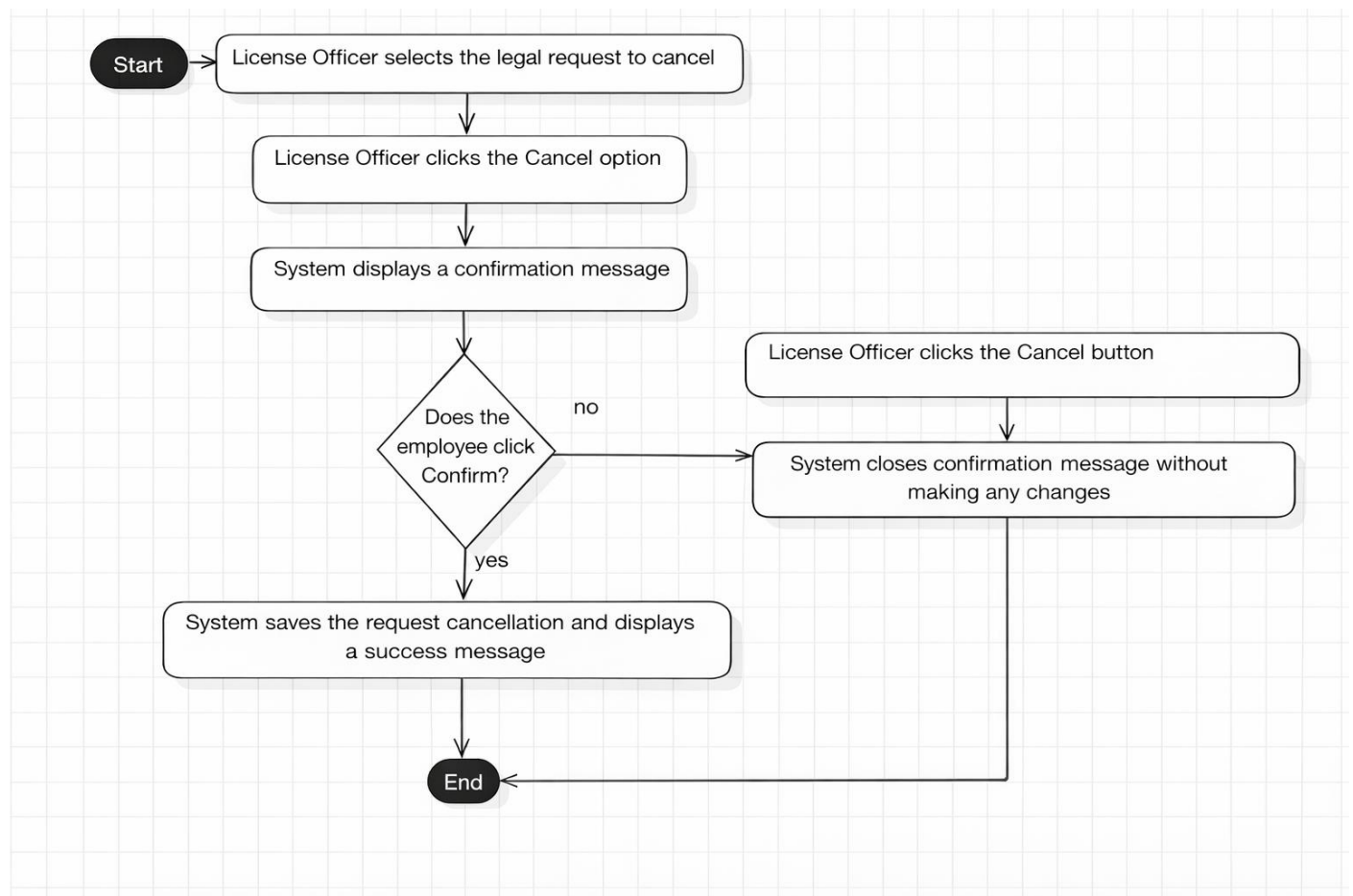
Description of the requirement to cancelled application

Use Case Title	Cancel Application
Actors	License Officer
Preconditions	Login
Main Flow	1. The Employee opens the Manage Applications screen. 2. The Employee selects the target application from the list. 3. The Employee clicks the Cancel Application button. 4. The system displays a confirmation message. 5. The Employee clicks the Confirm button. 6. The system records the Last Edit Date and displays a success message.
Alternative Flow	During Step 4: The Employee declines the confirmation 1. The system displays the cancellation confirmation message. 2. The Employee selects Cancel/No. 3. The system closes the confirmation message without changing the application status.
Postcondition	None

Sequence diagram to cancelled application:



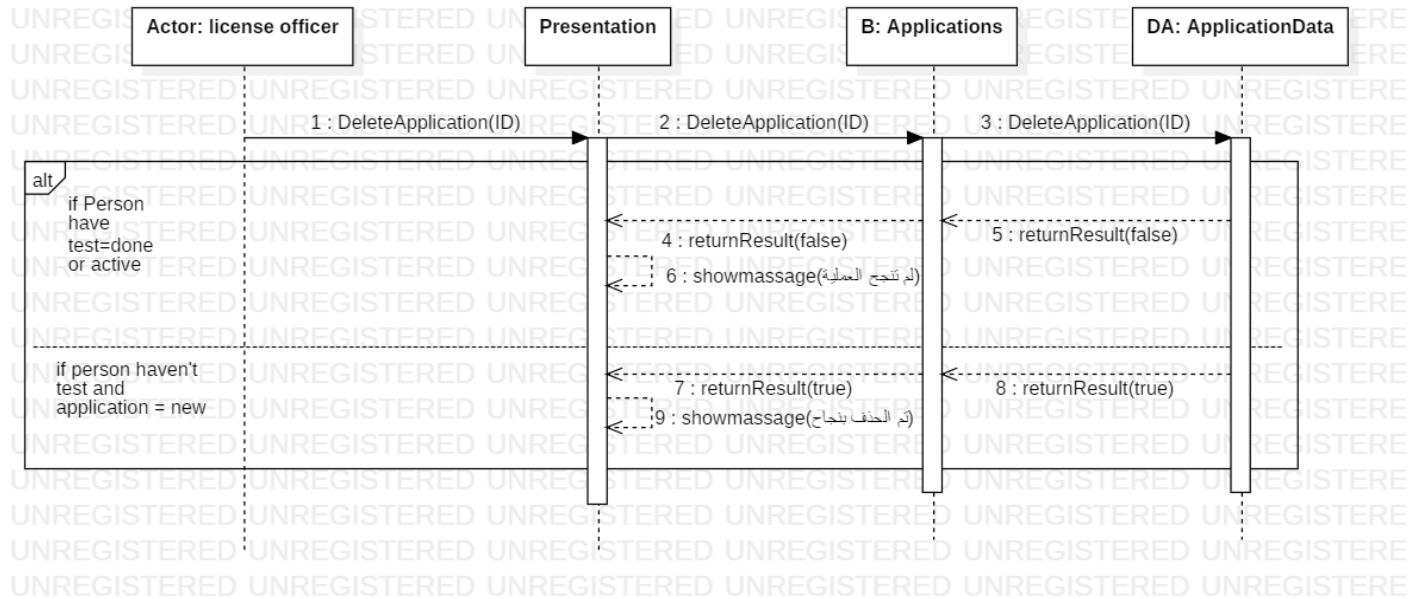
Activity Diagram cancelled application



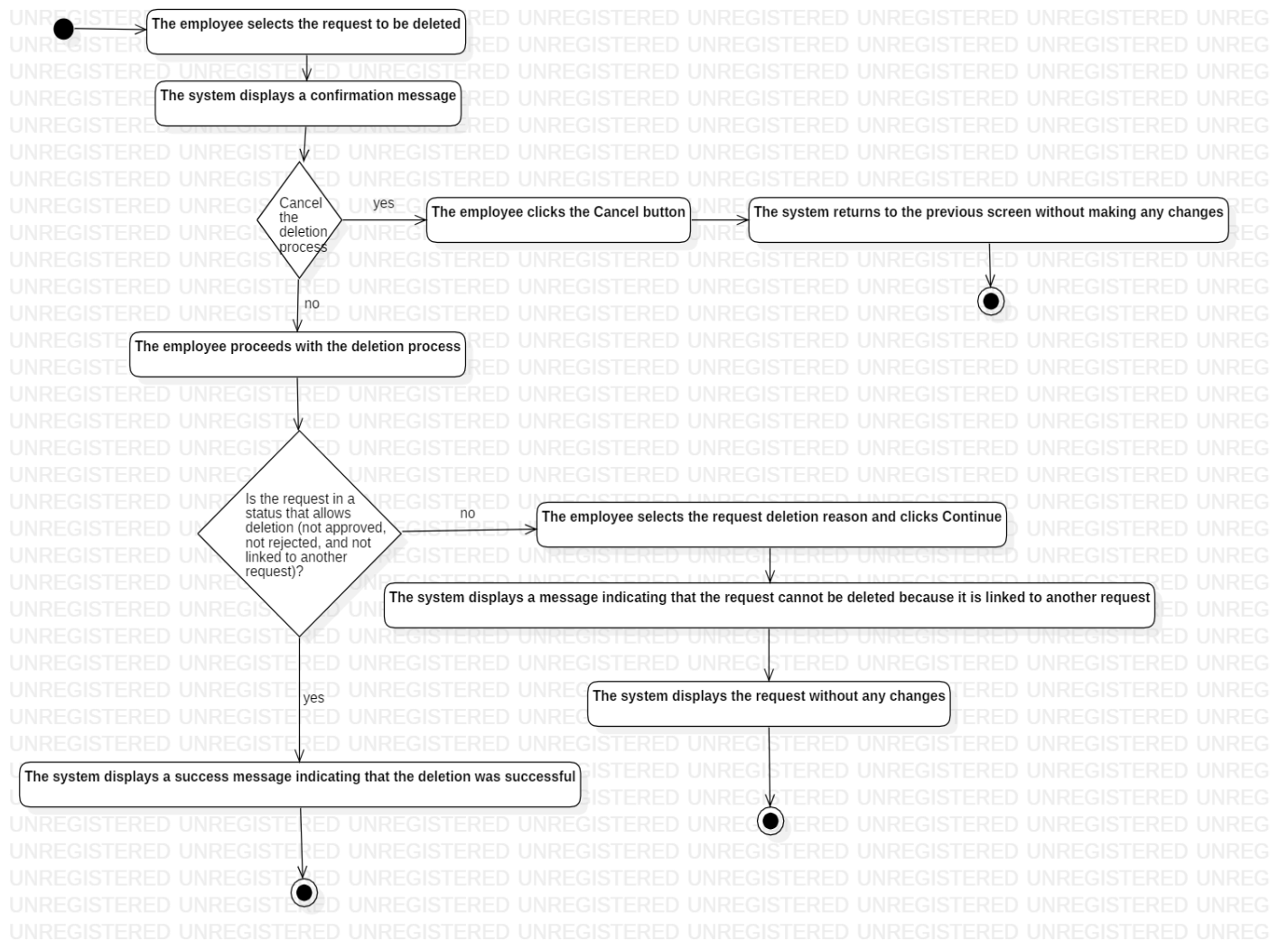
Description of the requirement to Delete application

Use Case Title	Delete Application
Actors	License Officer
Preconditions	Login
Main Flow	1. The Employee opens the Manage Applications screen. 2. The Employee selects the application to be deleted and clicks the Delete button. 3. The system displays a confirmation message for the deletion. 4. The Employee confirms the deletion process. 5. The system verifies that the application status is "New" and that it is not linked to any tests. 6. The system displays a success message confirming the operation.
Alternative Flow	Cancel Deletion Process: 1. The Employee selects the Cancel button. 2. The system returns the Employee to the previous screen without changing any data. Application Linked to Tests: 1. The Employee attempts to delete an application that has an associated test or is already completed/finished. 2. The system displays an error message: "Cannot delete the application because it is linked to tests." 3. The system returns to the application view without any changes.
Postcondition	None

Sequence diagram delete application



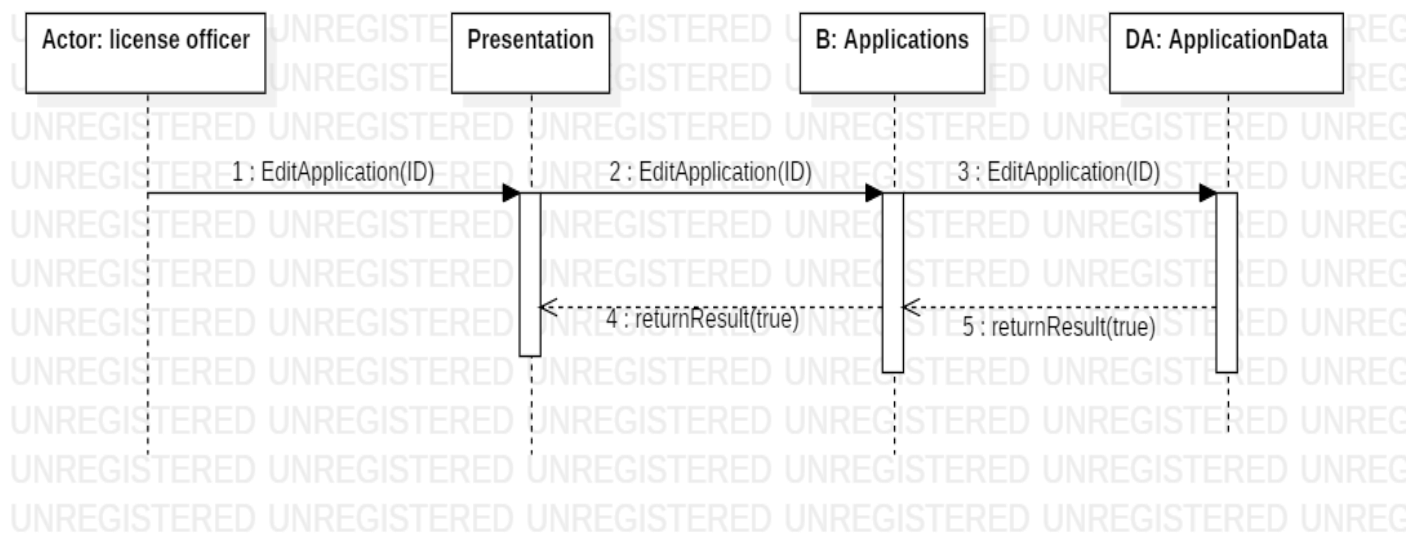
Activity Diagram delete application



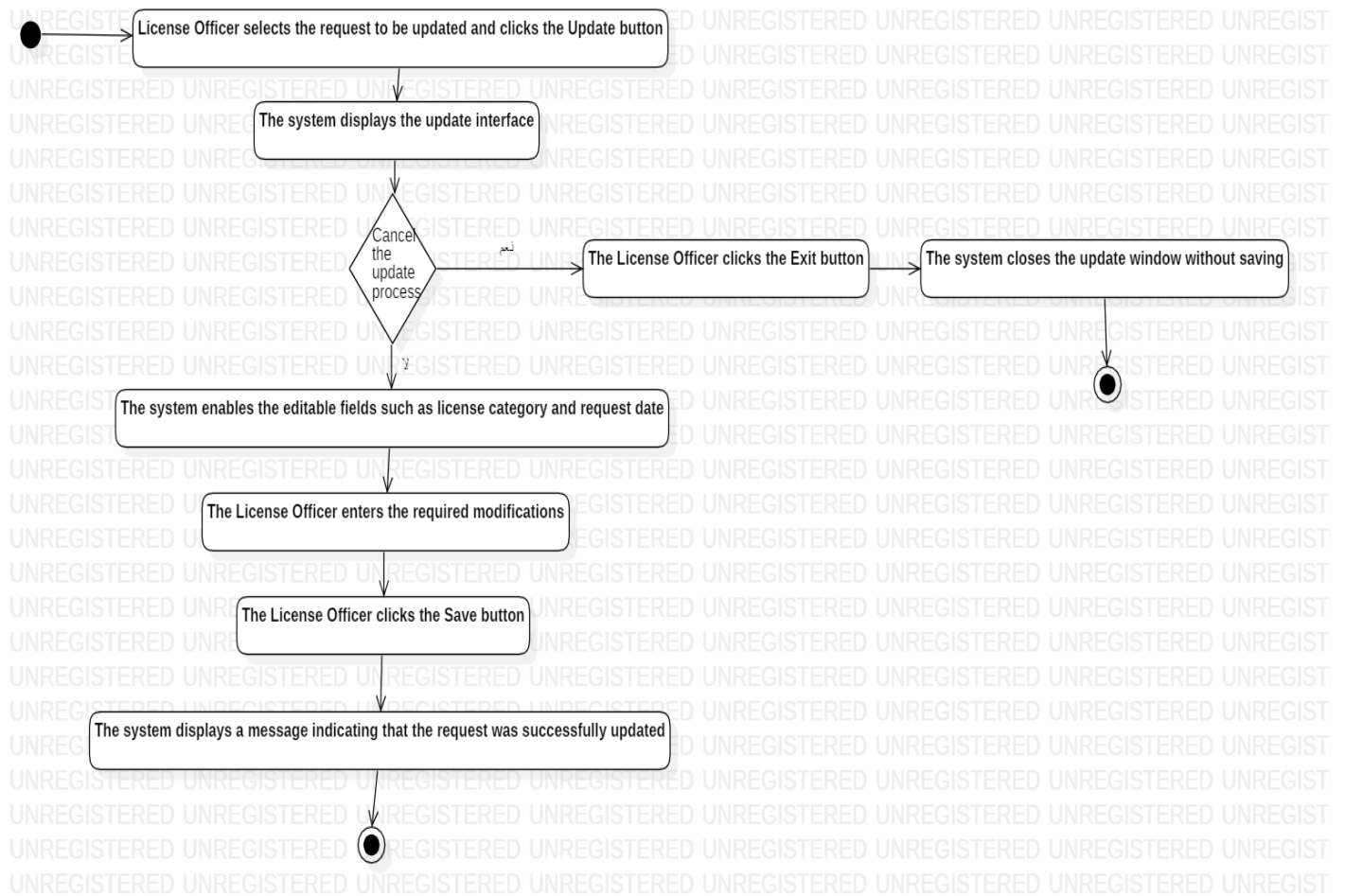
Description of the requirement to update application

Use Case Title	Update Application
Actors	License Officer
Preconditions	Login
Main Flow	1. The Employee opens the Manage Applications screen. 2. The Employee selects the target application to be updated and clicks Edit. 3. The system displays the Edit interface. 4. The system allows the Employee to update editable fields, such as License Class and Application Date. 5. The Employee enters the required modifications. 6. The Employee clicks the Save button. 7. The system displays a message: "Application updated successfully."
Alternative Flow	2. If the person already has an active application of the same type, the system displays an error message. 3. If the person already holds a license of the same class, the system displays an error message.
Postcondition	None

Sequence diagram update application



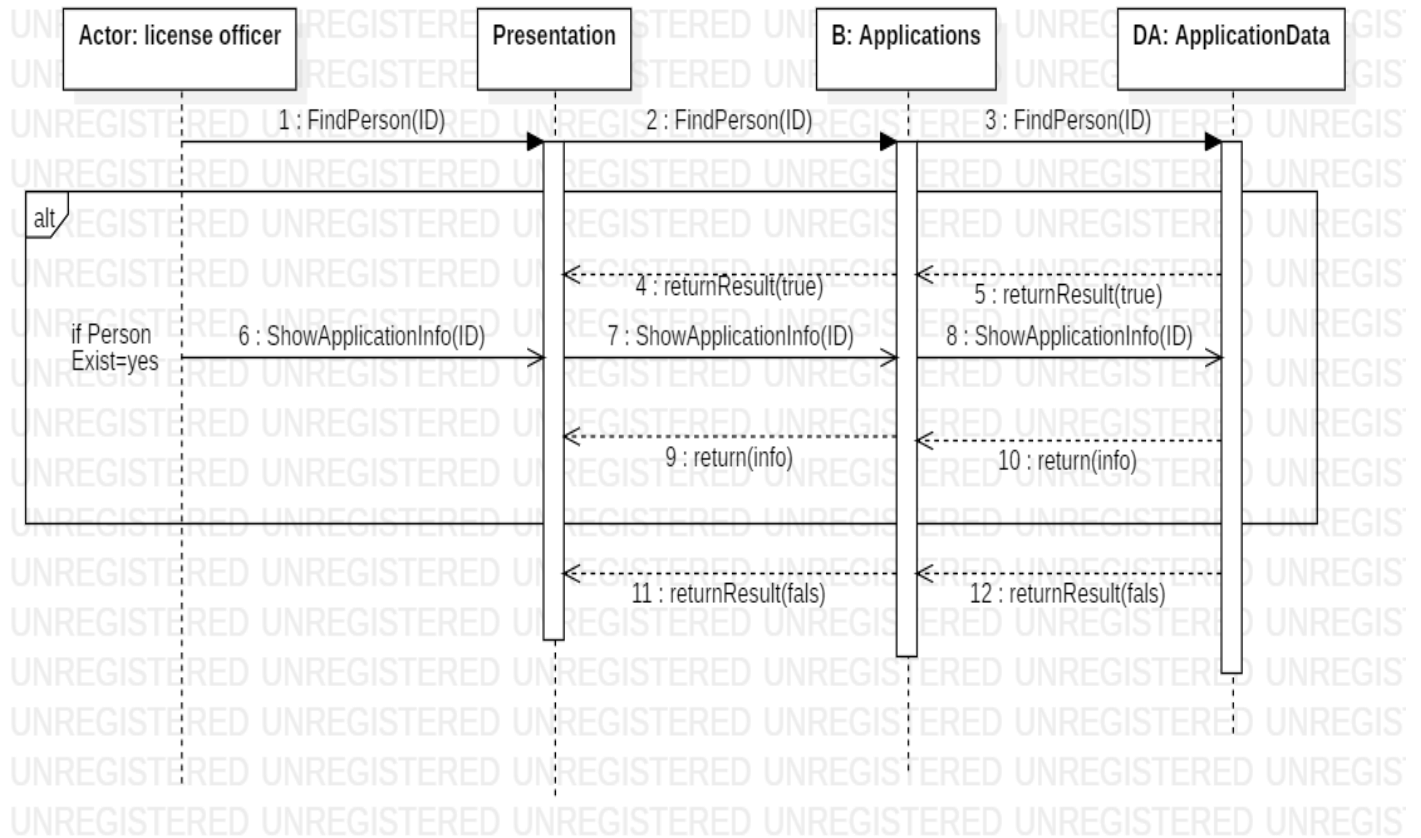
Activity Diagram update application



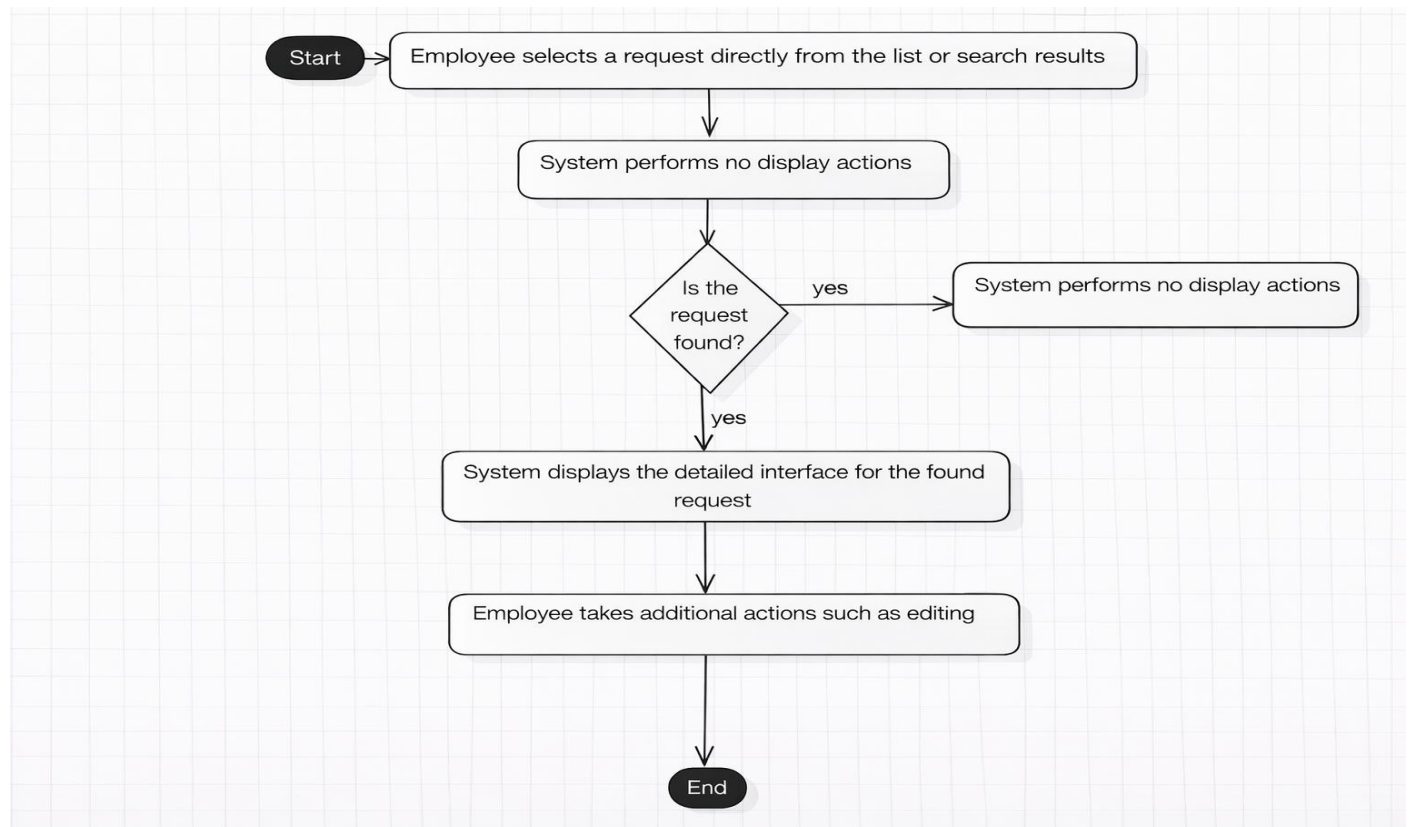
Description of the requirement to View application details

Use Case Title	View Application Details
Actors	License Officer
Preconditions	Login
Main Flow	1. The Employee opens the Manage Applications screen. 2. The Employee selects a specific application from the list or search results. 3. The system displays the details interface for the selected application. 4. The Employee can take additional actions, such as Update (Edit).
Alternative Flow	None
Postcondition	None

Sequence diagram View application details



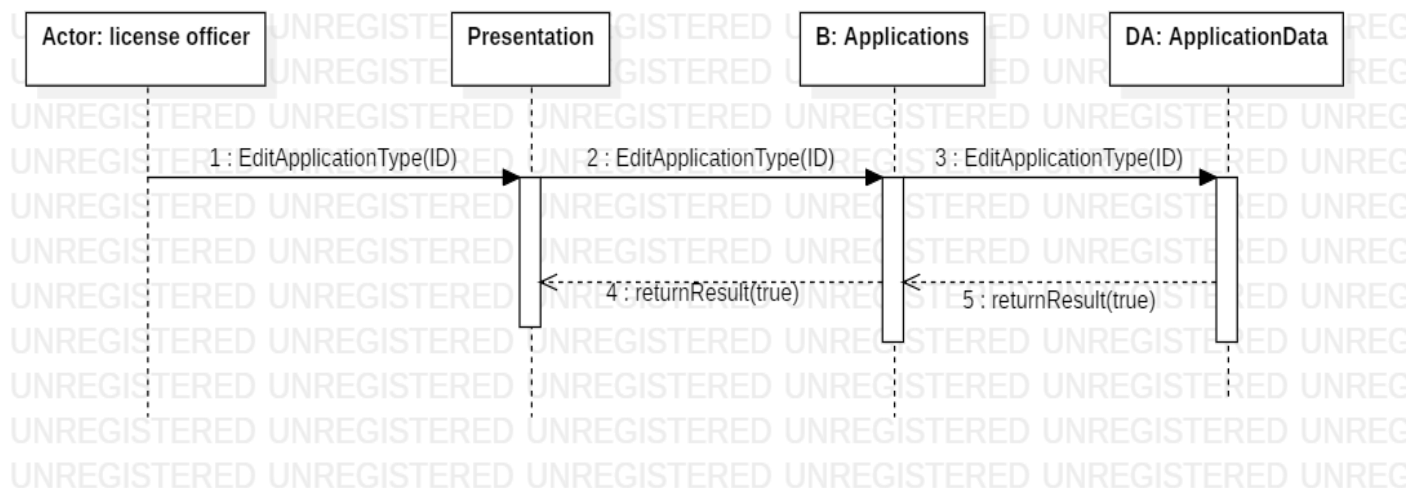
Activity Diagram View application details



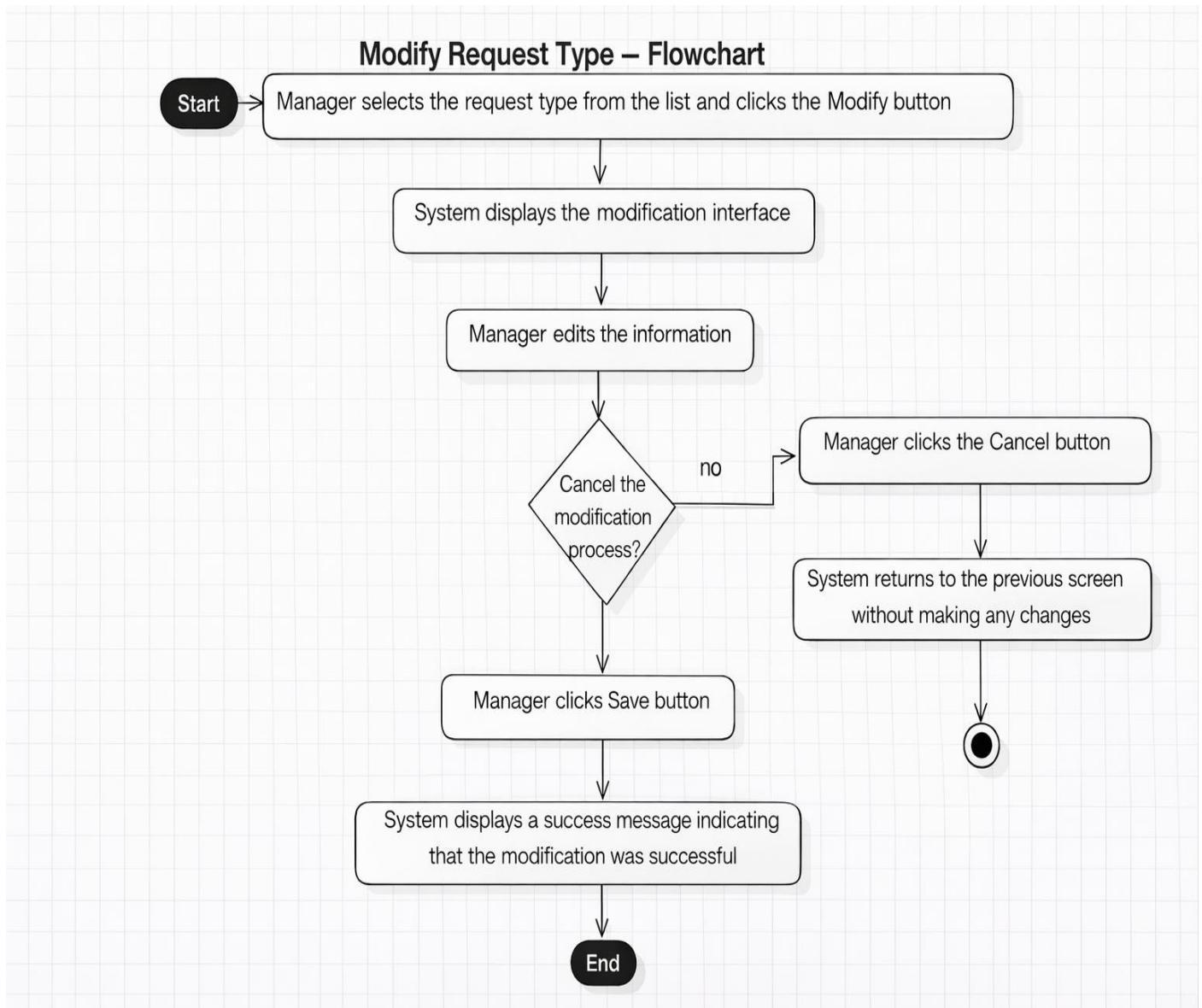
Description of the requirement to update application types

Use Case Title	Update Application Types
Actors	Admin
Preconditions	Login
Main Flow	1. The Admin opens the Manage Application Types screen. 2. The Admin selects an application type from the list and clicks the Edit button. 3. The system displays the Edit interface. 4. The Admin modifies the information. 5. The Admin clicks the Save button. 6. The system displays a message: "Application type updated successfully."
Alternative Flow	None
Postcondition	None

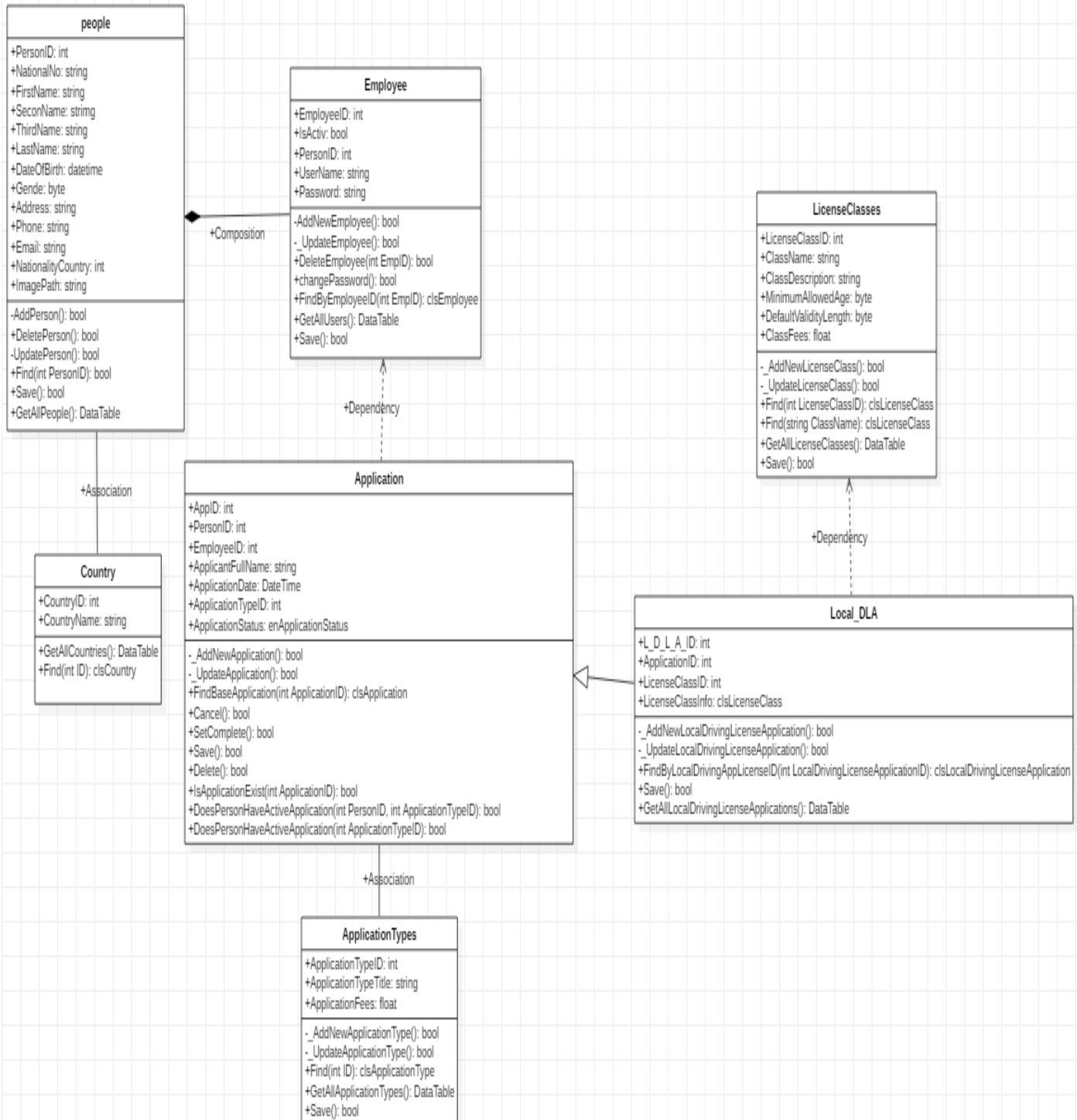
Sequence diagram update application types



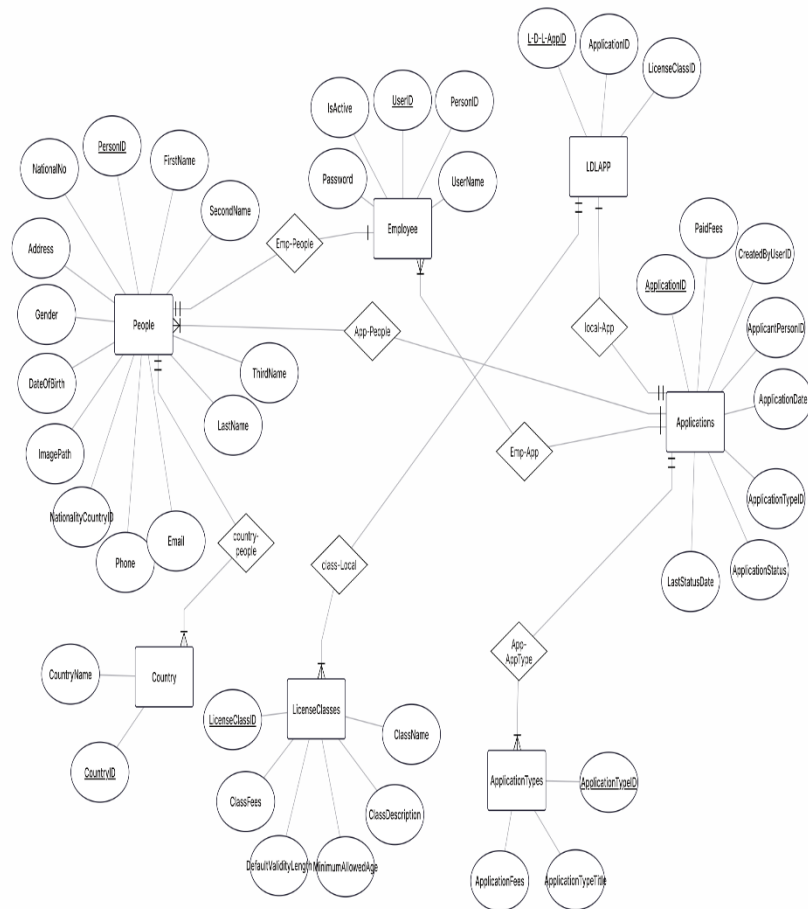
Activity Diagram update application types



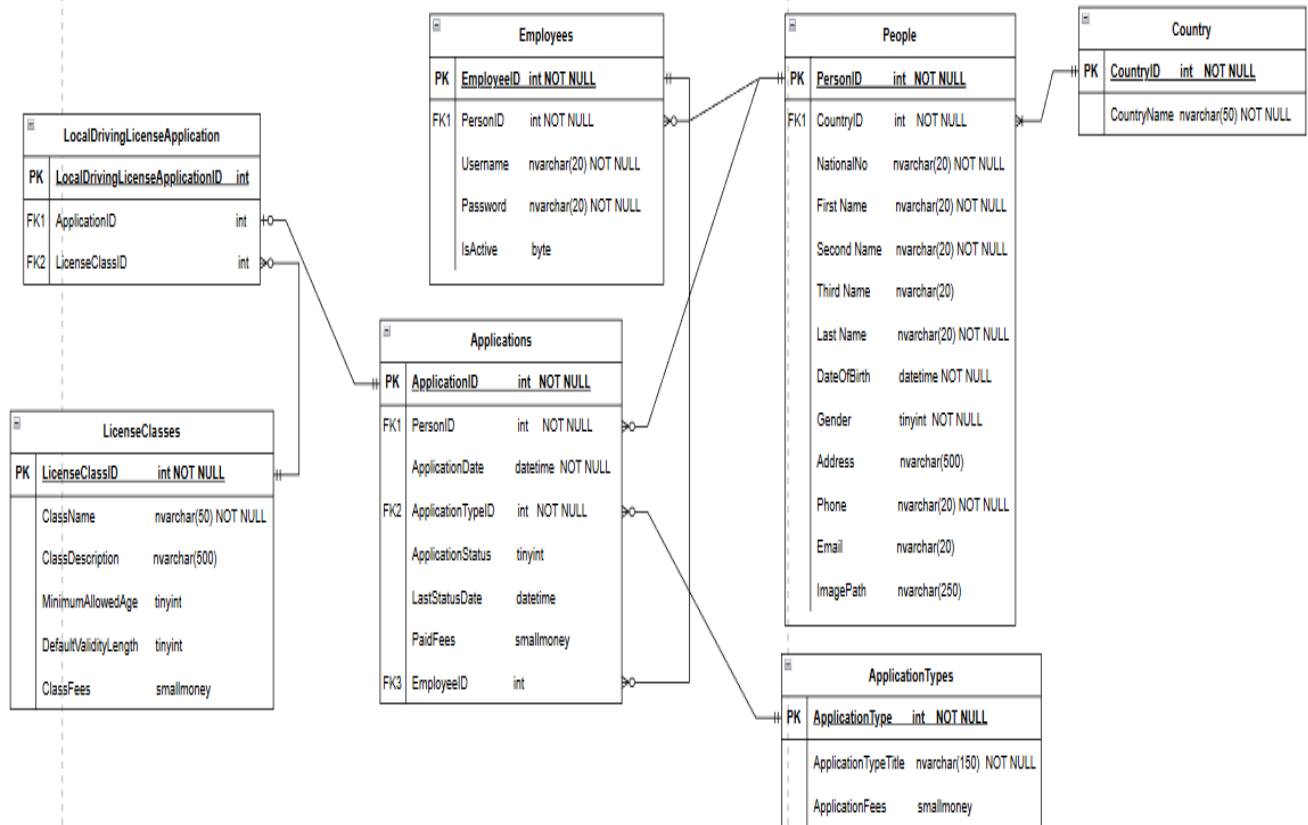
Analysis class diagram:



Entity Relationship Diagram ERD:



Schema Diagram :



5. Initial Test Cases:

ID	TC_01
Title	Entering a correct username and password.
Test Steps	1- Open the Login interface. 2- Enter the username and password. 3- Click the Login button.
Expected Result	Login successful.

ID	TC_02
Title	Entering an incorrect username or password.
Test Steps	1- Open the Login interface. 2- Enter the username and password. 3- Click the Login button.
Expected Result	A login failure message appears.

ID	TC_03
Title	Leaving the username or password field empty.
Test Steps	1- Open the Login interface. 2- Leave one of the two fields empty. 3- Click the Login button.
Expected Result	A login failure message appears due to an empty field.

ID	TC_04
Title	Logging in with an inactive account.
Test Steps	1- Open the Login interface. 2- Enter the username and password. 3- Click the Login button.
Expected Result	A login failure message appears because the account is inactive.

ID	TC_05
Title	Adding a person and filling in all data correctly.
Test Steps	1- Open the Add Person interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	Person added successfully.

ID	TC_06
Title	Entering a National No. that already exists while adding a person.
Test Steps	1- Open the Add Person interface 2- Fill in the data (using a duplicate National No.).
Expected Result	A notification appears stating that the National No. already exists.

ID	TC_07
Title	Entering an email with an incorrect format while adding a person.
Test Steps	1- Open the Add Person interface 2- Fill in the data .
Expected Result	A notification appears: "Please enter a valid email address."

ID	TC_08
Title	Leaving a required field empty while adding a person.
Test Steps	1- Open the Add Person interface. 2- Fill in the data (leaving at least one required field empty). 3- Click the Save button.
Expected Result	A message appears: "Please fill in all required fields."

ID	TC_09
Title	Deleting a person who has not made any applications.
Test Steps	1- Open the Manage People interface. 2- Select the target person. 3- Delete the person.
Expected Result	A success message appears.

ID	TC_10
Title	Deleting a person who has already made applications.
Test Steps	1- Open the Manage People interface. 2- Select the target person. 3- Delete the person.
Expected Result	A failure message appears (indicating the person cannot be deleted due to linked records).

ID	TC_11
Title	Creating an account for an employee already existing as a person in the system with correct data entry.
Test Steps	1- Open the Create Employee Account interface. 2- Search for the person. 3- Click the Next button. 4- Fill in the required account data. 5- Click the Save button.
Expected Result	A success message appears.

ID	TC_12
Title	Creating an account for an employee who does not exist as a person in the system.
Test Steps	1- Open the Create Employee Account interface. 2- Search for the person.
Expected Result	A message appears: "Person not found."

ID	TC_13
Title	Mismatch between password and confirmation during employee account creation.
Test Steps	1- Open the Create Employee Account interface. 2- Search for the person. 3- Click the Next button. 4- Fill in the data (entering different values in the password and confirmation fields).
Expected Result	A notification appears: "Passwords do not match."

ID	TC_14
Title	Deleting an employee who has not performed any tasks/actions.
Test Steps	1- Open the Manage Employees interface. 2- Select the target employee. 3- Delete the employee.
Expected Result	A success message appears.

ID	TC_15
Title	Changing an employee's password and entering the old password correctly.
Test Steps	1- Open the Change Password interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	A success message appears.

ID	TC_16
Title	Changing an employee's password and entering the old password incorrectly.
Test Steps	1- Open the Change Password interface. 2- Fill in the data (including an incorrect old password).
Expected Result	A notification appears: "Incorrect password."

ID	TC_17
Title	Creating a local license and selecting a category for a first-time applicant.
Test Steps	1- Open the Create Local License interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	A success message appears.

ID	TC_18
Title	Creating a local license and selecting a category for a person who already has an active application for the same category.
Test Steps	1- Open the Create Local License interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	A message appears: "You cannot submit an application because the person already has an active application."

ID	TC_19
Title	Creating a local license and selecting a category for a person who already holds a license in the same category.
Test Steps	1- Open the Create Local License interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	A message appears: "You cannot submit an application because the person already has a license in the same category."

ID	TC_20
Title	Deleting an application for a person who has not taken the tests yet.
Test Steps	1- Open the Manage Applications interface. 2- Select the target application. 3- Delete the application.
Expected Result	A success message appears.

ID	TC_21
Title	Deleting an application for a person after they have completed the tests.
Test Steps	1- Open the Manage Applications interface. 2- Select the target application. 3- Delete the application.
Expected Result	A failure message appears (indicating the application cannot be deleted because it is linked to existing test records).

ID	TC_22
Title	Canceling an application.
Test Steps	1- Open the Manage Applications interface. 2- Select the target application. 3- Click on Cancel Application .
Expected Result	A success message appears.

Initial RTM Requirements: Trackability Matrix

req-id	Use cases	analysis	System design	Detailed design	coding	Test cases
<u>RE-FR-PM-01</u>	<u>UC_01</u>	<u>PM-AN</u>				<u>TC_05</u> <u>TC_06</u> <u>TC_07</u> <u>TC_08</u>
<u>RE-FR-PM-02</u>	<u>UC_02</u>	<u>PM-AN</u>				<u>TC_09</u> <u>TC_10</u>
<u>RE-FR-PM-03</u>	<u>UC_03</u>	<u>PM-AN</u>				
<u>RE-FR-PM-04</u>	<u>UC_04</u>	<u>PM-AN</u>				
<u>RE-FR-PM-05</u>	<u>UC_05</u>	<u>PM-AN</u>				
<u>RE-FR-EM-06</u>	<u>UC_06</u>	<u>EM-AN</u>				<u>TC-11</u> <u>TC-12</u> <u>TC-13</u>
<u>RE-FR-EM-07</u>	<u>UC_07</u>	<u>EM-AN</u>				<u>TC-14</u>
<u>RE-FR-EM-08</u>	<u>UC_08</u>	<u>EM-AN</u>				
<u>RE-FR-EM-09</u>	<u>UC_09</u>	<u>EM-AN</u>				
<u>RE-FR-EM-10</u>	<u>UC_10</u>	<u>EM-AN</u>				
<u>RE-FR-EM-11</u>	<u>UC_11</u>	<u>EM-AN</u>				<u>TC-15</u> <u>TC-16</u>
<u>RE-FR-EM-12</u>	<u>UC_12</u>	<u>EM-AN</u>				<u>TC-01</u> <u>TC-02</u> <u>TC-03</u> <u>TC-04</u>
<u>RE-FR-EM-13</u>	<u>UC_13</u>	<u>EM-AN</u>				

<u>RE-FR-EM-14</u>	<u>UC 14</u>	EM-AN				
<u>RE-FR-EM-15</u>	<u>UC 15</u>	EM-AN				
<u>RE-FR-AM-16</u>	<u>UC 16</u>	<u>AM-AN</u>				TC-17 TC-18 TC-19
<u>RE-FR-AM-17</u>	<u>UC 17</u>	<u>AM-AN</u>				TC-22
<u>RE-FR-AM-18</u>	<u>UC 18</u>	<u>AM-AN</u>				TC-20 TC-21
<u>RE-FR-AM-19</u>	<u>UC 19</u>	<u>AM-AN</u>				
<u>RE-FR-AM-20</u>	<u>UC 20</u>	<u>AM-AN</u>				
<u>RE-FR-AM-21</u>	<u>UC 21</u>	<u>AM-AN</u>				
<u>RE-FR-AM-22</u>	<u>UC 22</u>	AM-AN				
<u>RE-FR-AM-23</u>	<u>UC 23</u>	AM-AN				

Chapter 5 System Design

1. Introduction:

This chapter offers a comprehensive overview of the design phase steps, encompassing system design and architecture determination. It underscores the importance of these steps in creating a robust system for collecting, analyzing, and utilizing user feedback to continuously enhance the software system.

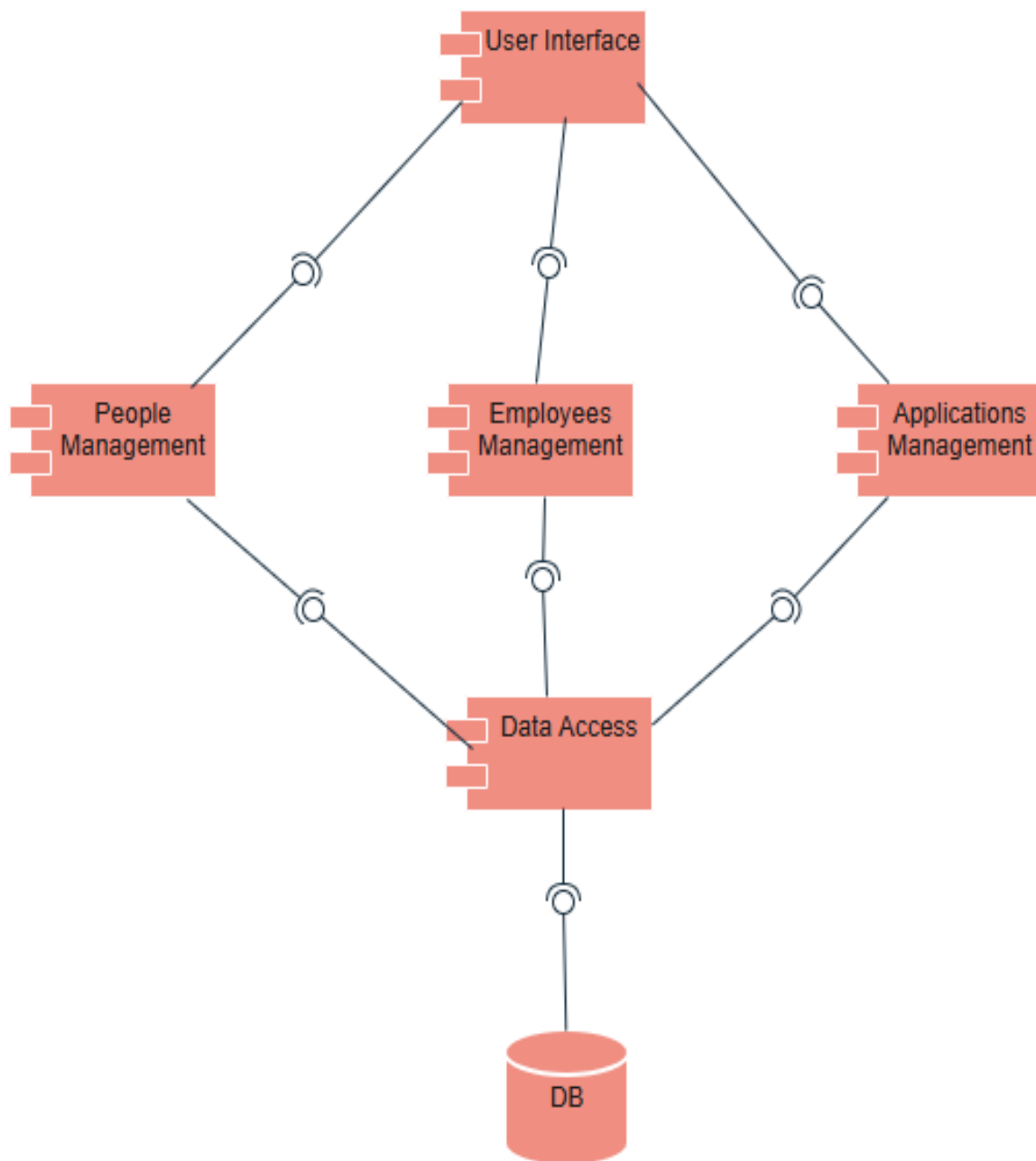
2. System Architecture:

In this section we will discuss system design where system design involves defining the overall structure and architecture of the software, ensuring that all components work seamlessly together. This phase establishes the blueprint for efficient data flow, interaction, and integration within the system.

The first constraints to take into consideration when building an architecture for a software system are **the non-functional requirements** (quality attributes), so first we will start with the non-functional requirements of our system

1. **Performance:** *Any license query must be completed in the shortest possible time.*
2. **Security:** Access permissions must be controlled based on roles (Role-Based Access Control).
3. **Maintainability:** The system architecture should support modifying or updating specific modules without impacting the entire system. Comprehensive documentation must be provided for all software components to facilitate future maintenance.
4. **Usability:** The user interface should be intuitive and easy to use for all user groups.

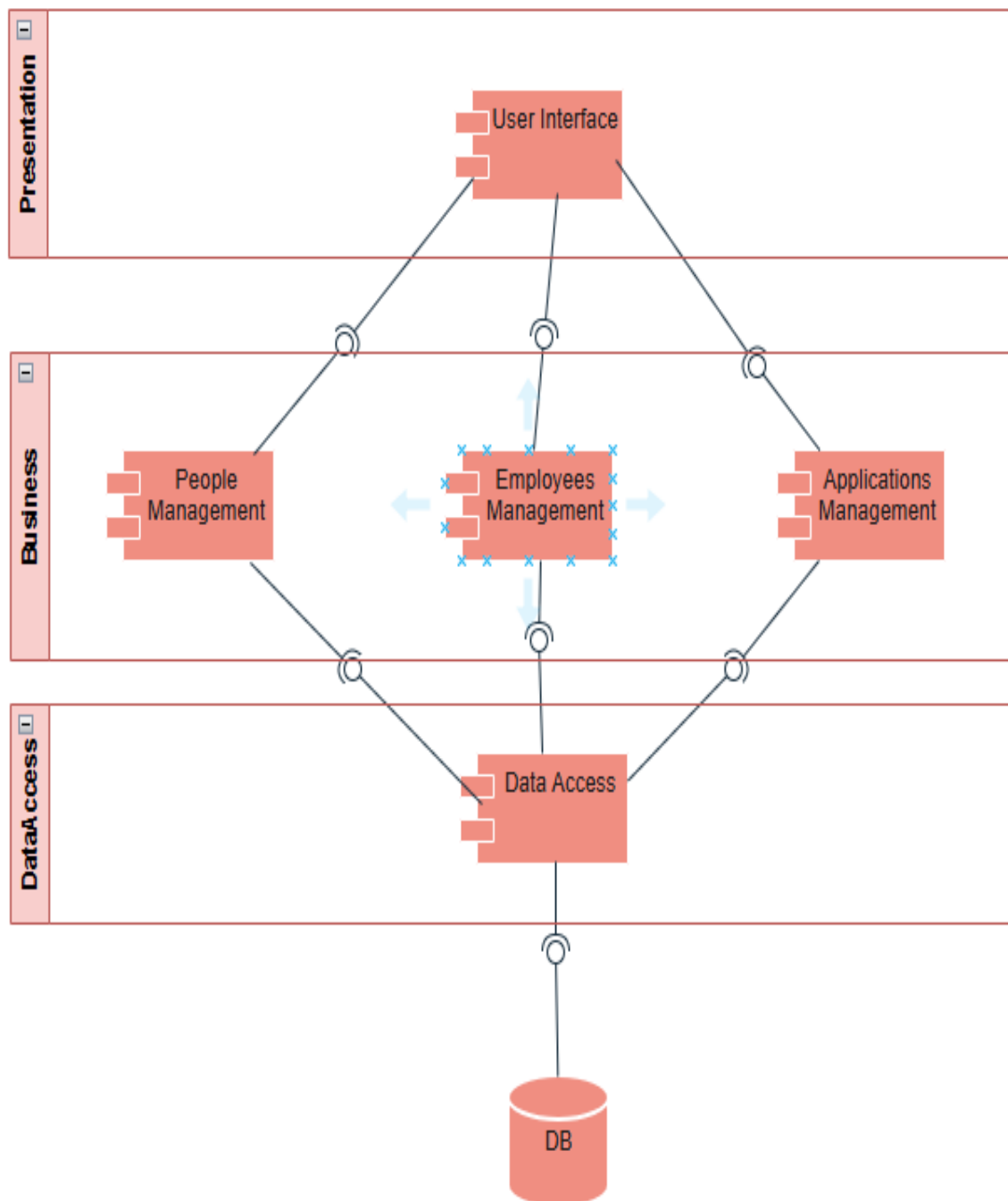
System decomposition - Component diagram:



Components functionalities:

1. **User Interface:** Responsible for enabling user interaction with the system. It presents forms, captures input, and displays results in a clear and intuitive manner.
2. **People Management:** Manages the storage, updating, and retrieval of person-related data within the system, such as names, identification, and personal information.
3. **Employees Management:** Handles account creation, login functionality, and assignment of user access rights based on predefined roles within the system.
4. **Applications Management:** Processes, tracks, and manages applications submitted by users, such as license requests or data update requests.
5. **Data Access:** Serves as the intermediary layer responsible for reading, updating, and managing data transactions between business logic components and the database securely and efficiently.
6. **Database:** The central storage system containing all data related to users, persons, and applications in a structured and persistent format.

System Architecture:



We used a layered architecture because it is suitable for our non-functional requirements.

Presentation Layer: This is the front-end layer that interacts directly with users. It handles input and output, displaying data in a user-friendly format.

Business Layer: Responsible for applying business rules and logic. It processes user input, manages workflows, and controls data flow between other layers.

Data Access Layer: Acts as the bridge to the database. It performs all read/write operations, ensuring secure and efficient communication with data storage.

4. Updating the RTM:

req-id	Use cases	analysis	System design	Detailed design	coding	Test cases
<u>RE-FR-PM-01</u>	<u>UC 01</u>	<u>PM-AN</u>	<u>AM-DL-01</u>			<u>TC 05</u> <u>TC 06</u> <u>TC 07</u> <u>TC 08</u>
<u>RE-FR-PM-02</u>	<u>UC 02</u>	<u>PM-AN</u>	<u>AM-DL-01</u>			<u>TC 09</u> <u>TC 10</u>
<u>RE-FR-PM -03</u>	<u>UC 03</u>	<u>PM-AN</u>	<u>AM-DL-01</u>			
<u>RE-FR-PM -04</u>	<u>UC 04</u>	<u>PM-AN</u>	AM-DL-01			
<u>RE-FR-PM -05</u>	<u>UC 05</u>	<u>PM-AN</u>	AM-DL-01			
<u>RE-FR-EM -06</u>	<u>UC 06</u>	<u>EM-AN</u>	AM-DL-01			<u>TC-11</u> <u>TC-12</u> <u>TC-13</u>
<u>RE-FR-EM -07</u>	<u>UC 07</u>	<u>EM-AN</u>	AM-DL-01			<u>TC-14</u>
<u>RE-FR-EM -08</u>	<u>UC 08</u>	<u>EM-AN</u>	AM-DL-01			
<u>RE-FR-EM -09</u>	<u>UC 09</u>	<u>EM-AN</u>	AM-DL-01			
<u>RE-FR-EM -10</u>	<u>UC 10</u>	<u>EM-AN</u>	AM-DL-01			
<u>RE-FR-EM -11</u>	<u>UC 11</u>	<u>EM-AN</u>	AM-DL-01			<u>TC-15</u> <u>TC-16</u>
<u>RE-FR-EM -12</u>	<u>UC 12</u>	<u>EM-AN</u>				<u>TC-01</u> <u>TC-02</u> <u>TC-03</u> <u>TC-04</u>
<u>RE-FR-EM -13</u>	<u>UC 13</u>	<u>EM-AN</u>	AM-DL-01			

<u>RE-FR-EM-14</u>	<u>UC 14</u>	EM-AN	AM-DL-01			
<u>RE-FR-EM-15</u>	<u>UC 15</u>	EM-AN	AM-DL-01			
<u>RE-FR-AM-16</u>	<u>UC 16</u>	<u>AM-AN</u>	AM-DL-01			<u>TC-17</u> <u>TC-18</u> <u>TC-19</u>
<u>RE-FR-AM-17</u>	<u>UC 17</u>	<u>AM-AN</u>	AM-DL-01			<u>TC-22</u>
<u>RE-FR-AM-18</u>	<u>UC 18</u>	<u>AM-AN</u>	AM-DL-01			<u>TC-20</u> <u>TC-21</u>
<u>RE-FR-AM-19</u>	<u>UC 19</u>	<u>AM-AN</u>	AM-DL-01			
<u>RE-FR-AM-20</u>	<u>UC 20</u>	<u>AM-AN</u>	AM-DL-01			
<u>RE-FR-AM-21</u>	<u>UC 21</u>	<u>AM-AN</u>	AM-DL-01			
<u>RE-FR-AM-22</u>	<u>UC 22</u>	AM-AN	AM-DL-01			
<u>RE-FR-AM-23</u>	<u>UC 23</u>	AM-AN	AM-DL-01			

Chapter 6 Practical Implementation

1. Introduction:

In this chapter, we present the practical implementation of the Driving License Management System. We detail the development environment, programming language, and supporting technologies used in building the system. The chapter also includes screenshots of the system interfaces that demonstrate key functionalities such as adding users, issuing licenses, and managing trainee records. Finally, we execute a set of predefined test cases to verify the system's behavior and ensure it meets the specified functional requirements under various operational scenarios.

2. Used Tools:

- Application Type: *Desktop Application*
- Programming Language: *C#*
- UI Framework: *Windows Forms*
- Development Environment: *Visual Studio*
- Database: *SQL Server or SQLite*
- Supporting Libraries: *Entity Framework, RDLC Reports*

3. System Interfaces:

log in interface:



**WELCOM TO
DRIVING & VEHICLE
LICENSE DEPARTMENT
(DVLD) SYSTEM**



Login to your account

Username:  Hasan

Password:  ****

☒ Remember Me.

 Login

Main Screen:



Person information :

Person ID : 1036

Name :  hasan mhdali hamdan alhoms

National No :  n6 Date Of Birth :  5/1/2003 12:00:00 AM

Gender :  Male Phone :  0923462426

Email :  hasan@gmail.com

Address :  midan Country :  Syria

[Edit Person Info](#)



Login Information :

User ID : 21 User Name : Hasoun1 Is Active : Yes

 Close

new local driving license:



Local Driving License Applications

Filter By: None

L.D.L.AppID	Driving Class
39	Class 4 - Co

Records: 1

New Local Driving License Application

New Local Driving License Application

Personal Info

Application Info

D.L.Application ID:

Application Date:

License Class:

Application Fees:

Created By:

[??]

1/1/2026

Class 5 - Agricultural

15

Hasan

Passed Tests

Status

0

New

Close

4. Test Cases Execution:

ID	TC_01
Title	Entering a correct username and password.
Test Steps	1- Open the Login interface. 2- Enter the username and password. 3- Click the Login button.
Expected Result	Login successful.
Actual Result	Login successful.
Pass/Fail	Pass

ID	TC_02
Title	Entering an incorrect username or password.
Test Steps	1- Open the Login interface. 2- Enter the username and password. 3- Click the Login button.
Expected Result	A login failure message appears.
Actual Result	A login failure message appears.
Pass/Fail	Pass

ID	TC_03
Title	Leaving the username or password field empty.
Test Steps	1- Open the Login interface. 2- Leave one of the two fields empty. 3- Click the Login button.
Expected Result	A login failure message appears due to an empty field.
Actual Result	A login failure message appears due to an empty field.
Pass/Fail	Pass

ID	TC_04
Title	Logging in with an inactive account.
Test Steps	1- Open the Login interface. 2- Enter the username and password. 3- Click the Login button.
Expected Result	A login failure message appears because the account is inactive.
Actual Result	A login failure message appears because the account is inactive.
Pass/Fail	Pass

ID	TC_05
Title	Adding a person and filling in all data correctly.
Test Steps	1- Open the Add Person interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	Person added successfully.
Actual Result	Person added successfully.
Pass/Fail	Pass

ID	TC_06
Title	Entering a National No. that already exists while adding a person.
Test Steps	1- Open the Add Person interface 2- Fill in the data (using a duplicate National No.).
Expected Result	A notification appears stating that the National No. already exists.
Actual Result	A notification appears stating that the National No. already exists.
Pass/Fail	Pass

ID	TC_07
Title	Entering an email with an incorrect format while adding a person.
Test Steps	1- Open the Add Person interface 2- Fill in the data .
Expected Result	A notification appears: "Please enter a valid email address."
Actual Result	A notification appears: "Please enter a valid email address."
Pass/Fail	Pass

ID	TC_08
Title	Leaving a required field empty while adding a person.
Test Steps	1- Open the Add Person interface. 2- Fill in the data (leaving at least one required field empty). 3- Click the Save button.
Expected Result	A message appears: "Please fill in all required fields."
Actual Result	A message appears: "Please fill in all required fields."
Pass/Fail	Pass

ID	TC_09
Title	Deleting a person who has not made any applications.
Test Steps	1- Open the Manage People interface. 2- Select the target person. 3- Delete the person.
Expected Result	A success message appears.
Actual Result	A success message appears.
Pass/Fail	Pass

ID	TC_10
Title	Deleting a person who has already made applications.
Test Steps	1- Open the Manage People interface. 2- Select the target person. 3- Delete the person.
Expected Result	A failure message appears (indicating the person cannot be deleted due to linked records).
Actual Result	A failure message appears (indicating the person cannot be deleted due to linked records).
Pass/Fail	Pass

ID	TC_11
Title	Creating an account for an employee already existing as a person in the system with correct data entry.
Test Steps	1- Open the Create Employee Account interface. 2- Search for the person. 3- Click the Next button. 4- Fill in the required account data. 5- Click the Save button.
Expected Result	A success message appears.
Actual Result	A success message appears.
Pass/Fail	Pass

ID	TC_12
Title	Creating an account for an employee who does not exist as a person in the system.
Test Steps	1- Open the Create Employee Account interface. 2- Search for the person.
Expected Result	A message appears: "Person not found."
Actual Result	A message appears: "Person not found."
Pass/Fail	Pass

ID	TC_13
Title	Mismatch between password and confirmation during employee account creation.
Test Steps	1- Open the Create Employee Account interface. 2- Search for the person. 3- Click the Next button. 4- Fill in the data (entering different values in the password and confirmation fields).
Expected Result	A notification appears: "Passwords do not match."
Actual Result	A notification appears: "Passwords do not match."
Pass/Fail	Pass

ID	TC_14
Title	Deleting an employee who has not performed any tasks/actions.
Test Steps	1- Open the Manage Employees interface. 2- Select the target employee. 3- Delete the employee.
Expected Result	A success message appears.
Actual Result	A success message appears.
Pass/Fail	Pass

ID	TC_15
Title	Changing an employee's password and entering the old password correctly.
Test Steps	1- Open the Change Password interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	A success message appears.
Actual Result	A success message appears.
Pass/Fail	Pass

ID	TC_16
Title	Changing an employee's password and entering the old password incorrectly.
Test Steps	1- Open the Change Password interface. 2- Fill in the data (including an incorrect old password).
Expected Result	A notification appears: "Incorrect password."
Actual Result	A notification appears: "Incorrect password."
Pass/Fail	Pass

ID	TC_17
Title	Creating a local license and selecting a category for a first-time applicant.
Test Steps	1- Open the Create Local License interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	A success message appears.
Actual Result	A success message appears.
Pass/Fail	Pass

ID	TC_18
Title	Creating a local license and selecting a category for a person who already has an active application for the same category.
Test Steps	1- Open the Create Local License interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	A message appears: "You cannot submit an application because the person already has an active application."
Actual Result	A message appears: "You cannot submit an application because the person already has an active application."
Pass/Fail	Pass

ID	TC_19
Title	Creating a local license and selecting a category for a person who already holds a license in the same category.
Test Steps	1- Open the Create Local License interface. 2- Fill in the required data. 3- Click the Save button.
Expected Result	A message appears: "You cannot submit an application because the person already has a license in the same category."
Actual Result	A message appears: "You cannot submit an application because the person already has a license in the same category."
Pass/Fail	Pass

ID	TC_20
Title	Deleting an application for a person who has not taken the tests yet.
Test Steps	1- Open the Manage Applications interface. 2- Select the target application. 3- Delete the application.
Expected Result	A success message appears.
Actual Result	A success message appears.
Pass/Fail	Pass

ID	TC_22
Title	Canceling an application.
Test Steps	1- Open the Manage Applications interface. 2- Select the target application. 3- Click on Cancel Application .
Expected Result	A success message appears.
Actual Result	A success message appears.
Pass/Fail	Pass

6. RTM Last Version:

req-id	Use cases	analysis	System design	Detailed design	coding	Test cases
<u>RE-FR-PM-01</u>	<u>UC_01</u>	<u>PM-AN</u>	<u>AM-DL-01</u>		source	<u>TC_05</u> <u>TC_06</u> <u>TC_07</u> TC_088
<u>RE-FR-PM-02</u>	<u>UC_02</u>	<u>PM-AN</u>	<u>AM-DL-01</u>		source	<u>TC_09</u> TC_10
<u>RE-FR-PM-03</u>	<u>UC_03</u>	<u>PM-AN</u>	<u>AM-DL-01</u>		source	
<u>RE-FR-PM-04</u>	<u>UC_04</u>	<u>PM-AN</u>	AM-DL-01		source	
<u>RE-FR-PM-05</u>	<u>UC_05</u>	<u>PM-AN</u>	AM-DL-01		source	
<u>RE-FR-EM-06</u>	<u>UC_06</u>	<u>EM-AN</u>	AM-DL-01		source	<u>TC-11</u> <u>TC-12</u> <u>TC-13</u>
<u>RE-FR-EM-07</u>	<u>UC_07</u>	<u>EM-AN</u>	AM-DL-01		source	<u>TC-14</u>
<u>RE-FR-EM-08</u>	<u>UC_08</u>	<u>EM-AN</u>	AM-DL-01		source	
<u>RE-FR-EM-09</u>	<u>UC_09</u>	<u>EM-AN</u>	AM-DL-01		source	
<u>RE-FR-EM-10</u>	<u>UC_10</u>	<u>EM-AN</u>	AM-DL-01		source	
<u>RE-FR-EM-11</u>	<u>UC_11</u>	<u>EM-AN</u>	AM-DL-01		source	<u>TC-15</u> <u>TC-16</u>
<u>RE-FR-EM-12</u>	<u>UC_12</u>	<u>EM-AN</u>				<u>TC-01</u> <u>TC-02</u> <u>TC-03</u> <u>TC-04</u>

<u>RE-FR-EM-13</u>	<u>UC 13</u>	EM-AN	AM-DL-01		source	
<u>RE-FR-EM-14</u>	<u>UC 14</u>	EM-AN	AM-DL-01		source	
<u>RE-FR-EM-15</u>	<u>UC 15</u>	EM-AN	AM-DL-01		source	
<u>RE-FR-AM-16</u>	<u>UC 16</u>	<u>AM-AN</u>	AM-DL-01		source	TC-17 TC-18 TC-19
<u>RE-FR-AM-17</u>	<u>UC 17</u>	<u>AM-AN</u>	AM-DL-01		source	TC-22
<u>RE-FR-AM-18</u>	<u>UC 18</u>	<u>AM-AN</u>	AM-DL-01		source	TC-20 TC-21
<u>RE-FR-AM-19</u>	<u>UC 19</u>	<u>AM-AN</u>	AM-DL-01		source	
<u>RE-FR-AM-20</u>	<u>UC 20</u>	<u>AM-AN</u>	AM-DL-01		source	
<u>RE-FR-AM-21</u>	<u>UC 21</u>	<u>AM-AN</u>	AM-DL-01		source	
<u>RE-FR-AM-22</u>	<u>UC 22</u>	AM-AN	AM-DL-01		source	
<u>RE-FR-AM-23</u>	<u>UC 23</u>	AM-AN	AM-DL-01		source	

Chapter7 Conclusion

1. Introduction:

As a result of this project, we developed an integrated software system for managing driving license services in an electronic and organized manner. The system aims to improve the efficiency and accuracy of handling driving license requests by enabling individuals to submit and track their applications electronically, while allowing employees and administrators to manage persons, users, requests, tests, and licenses through a centralized platform.

Throughout the project, modern software engineering practices were applied, including requirements engineering, UML modeling, and Agile Scrum methodology. These practices helped ensure that the system requirements were well-defined, traceable, and aligned with real-world driving license management processes.

The developed system enhances service transparency, reduces manual effort, minimizes errors, and improves overall user experience for both citizens and system staff. This project demonstrates how applying structured software engineering techniques can result in a reliable and scalable information system suitable for real governmental environments.

2. References

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- <https://www.drawio.com/>
- <https://www.scrumguides.org/>
- <https://www.erdplus.com/>